GROUP ASSIGNMENT

TECHNOLOGY PARK MALAYSIA

NP3F2509IT(CE)

INVESTIGATION REPORT

EventNexus: AI-Enabled Secure Cloud-Based Event Management System for Multi-Organization Collaboration

HAND OUT DATE: 29st April, 2026

HAND IN DATE: 12th January, 2026

WEIGHTAGE: 50%

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Abstract

It includes the design and research behind the EventNexus - a secure AI-enabled, cloud-based solution for events, developed with many organizations. Event management platforms are usually plagued by the same old issues: lack of proper automation, weak links in organization collaboration, poor security protocols, and the ever-scaling problem. However, EventNexus is an intelligent platform equipped with Artificial Intelligence, Cloud Computing, and Role-Based Access Control to better plan, manage, and engage participants in events. This paper has reviewed literature working on AI in event management and cloud-based systems, security in multi-tenant environments, and current-day platforms like Eventbrite, Cvent, and Whova. The analysis includes current solutions' deficits that need contesting with new proposed solutions that may offer more security and collaboration.

The technology involved will be the MERN Stack, MongoDB Database, AWS Cloud Infrastructure, JWT/OAuth 2.0 authentication, AI technologies for predictive analysis, personalized event recommendation, chatbot for customer support, QR code-based ticketing, and real-time dashboards.

Working in a Waterfall-Prototyping hybrid model, the methodology on one side maintained structured development and on the other side allowed any iterative user feedback.

The survey was set out to perform primary data collection that captures the user need and through validation, the system requirements. In fact, there are very high demands with respect to secure login, cloud storage, AI-based recommendations, reminders, and multi-organizational collaboration.

The report suggests a solution designated to be called EventNexus, which, according to the report, is a scalable and workable solution for the modern event management problem. It will provide a good technical and academic base for further development and realization in the future.

Acknowledgement

I extend my sincere words of appreciation to each one of you for having supported and guided me through the completion of this investigation report.

At first, I would like to express my deepest gratitude to Dr. R. N. Thakur, Module Leader, for the input in academic guidance and feedback that added value to this project. Encouragement from him was an impetus for the journey towards the realization of the project.

I would like to express my sincere gratitude to Er. Shambhu Gautam for guiding me through the whole process of investigation and research. He advised much expertise, provided many insightful comments and suggestions. His supervision has also made me understand which way to follow and what quality this work is supposed to have.

Special thanks to Lord Buddha Education Foundation (LBEF College) for creating an ambience of encouragement in learning and offering all the academic resources required in carrying out this study.

Finally, I would like to thank Asia Pacific University (APU) who has provided me with an academic framework that has allowed me very productive possibilities to move this project toward being a success.

I would also thank my friends, participants, and virtually all other people that stepped directly or indirectly toward the successful completion of this inquiry report.

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Chapter 1: Introduction to the Study

Background of the Project

Digital technologies have been scaling up at a very high pace over the past years. As a consequence, the mode of planning, coordinating, and delivering events by organizations has transformed. This has been the revolution for the events industry in the last decade - paper-based and manual in support of small corporate events to cloud-based and data-driven in support of large global conferences. That has largely been possible due to the integration of two big contributing factors. On one hand, cloud computing is all about releasing infrastructure as a service on-demand and marketing products globally. On the other hand, it's the evolution of Artificial Intelligence technologies that enable smart automation, a personalized user experience, and predictive decision making.

Now, events are more and more part of the common effort of many organizations. Therefore, it is very frequent that one is attending a conference, an exhibition, a workshop where academia, industry, and government agencies are co-hosts. However, while most of the public platforms provided in this digital era for inter-organizational collaboration apply siloed systems, smart automation, proper security, and multi-business collaboration are often lacking. It has been observed from research that current means of event management require extensive individual effort for key administrative burdens and a host of other related issues, many of which include modern software solutions related to ticketing, event scheduling, and volunteering management (Hamid & Halim, The Transformative Role of Artificial Intelligence in the Event Management Industry, 2023). However, other projections released from the AI in Event Management Market analysis for 2024 warned that the global AI-based event platform market might experience exponential growth, reaching USD 14.2 billion in 2033 from an estimated USD 1.8 billion in 2020, a signal staggering the industry toward smart cloud-based platforms.

EventNexus is a single window plus smart and secure online platform which shall integrate AI analytics, RBAC, powerful AWS cloud-based computing infrastructure to assist the solution from creating events to engaging and reviewing them post-execution.Problem Background

Despite giant strides taken technologically, several serious and constant problems remain which hamper effective event management, with an inter-company, multi-organizational environment in the head.

Lack of Intelligent Automation

Most of the current event management platforms come with straightforward functionalities for automation. Therefore, it hardly incorporates advanced functions of this nature: attendance prediction, recommendation personalization, and intelligent chatbot support. Literature systematic review in Event Management Journal points out that the employment of AI, namely through predictive analytics, chatbots, marketing automation, and real-time monitoring, has been successful in improving the processes under event management (Kumar & Ratten). These applications, generally, have not yet reached a level of diffusion. Without these capabilities, one tends to be dependent on subjective judgment and manual coordination done by the event planner, which increases the risk of operations mistakes and resource wastage.

Inefficient Multi-Organization Collaboration

It needs perfect coordination for co-hosted events among various categories of stakeholders: the administrator, organizer, and participants, who may even belong to different organizations with totally different workflows, policies, and data requirements. Right now, there is no support in the present platforms for structured, role-differentiated collaboration either. The lack of proper inter-organizational tools results in wasted inter-organizational communication time, duplicated effort, and squandered resources.

Data Security and Privacy Vulnerabilities

Most of the time, event management platforms are where a lot of sensitive data is stored or has to be maintained regarding user identities, payment credentials, and organizations' metadata. The studies on cloud architectures with multiple tenants go to support that co-located tenants in shared-cloud environments hold a potential high-risk factor for unauthorized access and data leakage in the absence of an applique mechanism for access control; these enablers become quite critical in the presence of located tenants (Alobaywi et al., 2026).

This further ensures that co-located tenants do not access others' data in case fine-grained permissions have to be extended; Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC) are very essential, as revealed by the research (Ratnam & Yasani, 2021).

Limited Scalability of Traditional Platforms

Traditional event management systems are not designed to scale elastically and dynamically, so as to cater to very high-demand spikes in users like those experienced during peak registration times. A dynamic, bursty workload, serverless and container-based cloud architectures would prove to be more elastic and cost-effective in comparison to conventional server deployments (Tari et al., 2024).

Lack of Integration with Modern Technologies

Most of today's solutions simply do not offer an intelligent mesh across modern-day cloud services, secured payment gateways, AI analytics engines, or robust RESTful APIs. The technocentric fractionation is creating friction for organizations toward delivering end-to-end seamless digital event experiences and structurally hindering the potential of the system to evolve in response to changing requirements.

Project Aim

This calls for the creation of 'EventNexus,' expressly designed bottom-up to offer event management solutions beyond the ordinary. EventNexus will, therefore, be broad-angled in its ability to serve different needs in modern, multi-organizational events. Cloud computing with AI merged into one using role-based access control are the perfect ingredients to prepare the most secure environment.In fact, the McKinsey team furthered that artificial intelligence can take over close to 70% of all tasks.

Well, that simply translates to more time being left in the free space for planners to consider valuable high-level tasks that can be strategic. Of course, the cloud-hosted infrastructure is particularly great when scaffolded on AWS, further ensuring that modern events get such on-demand scalability, high availability, and global access.This further gives stakeholders added assurance in the sense that they will be able to access information and functionalities based on the level of their roles - be it administrator, event organizer, or participant.

Having a structured access model is of prime importance to keep the integrity of the data intact in an environment hosting organization while being simultaneously compliant with modern-day standards for data protection, such as GDPR.

Potential Benefits

Tangible Benefits

Some of the key measurable operational metrics through which the proposed system will deliver an improvement are:

- Administration becoming easier and cheaper with AI-driven automation to schedule registration
- and communication tasks.
- More accurate predictions of the turnout for an event to enable optimization of the various
- resources which may be deployed, proper allocation of venue or staff, or even correct apportioning
- of food attendance.
- Secure digital ticketing with QR code verification, so no fraud or duplicate entries would be made
- Event organizers will need to schedule email notifications and reminders regarding their events
- The cloud scales elastically during peak loads so there is no downtime for critical registration events

Intangible Benefits

Long-term qualitative addition to the platform's value:

- Bettering participant satisfaction with personalized, AI-curated event recommendations that are keen
- on interests and inclinations
- Organizational trust reinforced based on clear security measures and practices

- A culture of cooperation between organizations, driven by access boundaries and shared
- infrastructure
- Data-driven decision-making to all levels of event management, beginning from strategic planning
- and going up to on-the-go operative adjustments
- Modern event management that would enrich an experience, making the organizing organizations feel that they are a part of an innovation-driven future

Target Users

The EventNexus platform has three main user groups:

- Administrators – He is tasked with the management of organization structures, user accounts, system settings, access management policies, and any kind of security surveillance. Administrators have control over all the tenant companies on the platform.
- Organizers – This is the role that is going to create, design, and manage events within his organization. Using AI-driven analytics, organizers will be able to coordinate participant registration, schedule sessions, and engage in automated communication.
- Attendees -These will be end-users who can find events of interest, register, and participate in them. Attendees will be able to be recommended specific events tailored to their needs with the use of AI, proceed to transact for these events in a very secure manner, receive QR unique digital tickets, and make opinions known after the event.

Scope and Objectives

Project Aim

This project basically targets at designing and developing a safe AI-driven cloud event management system that promotes collaboration between organizations to enhance operations and deliver an intelligent experience to the class of users in an extremely personal way.

Objectives

- To build a user-friendly and secure event management system seamlessly integrated in an AWS cloud.

- To provide AI-based features, such as event suggestions, predictions about attendees and chatbots.
- To provide data segregation for multiple tenants by using RBAC and IAM, which can help to maintain segregation between tenants (organisations).
- Assist the end users in terms of end-to-end event management, which involves event creation, registration and post-event analysis.
- To design a secure online payment system and a QR-based digital-ticketing system for the purpose of a seamless event verification from the attendees end.
- Administration and event-organisers will be offered analysis dashboards for event monitoring for fact-based decision making.

Scope

The key features that will be developed and provided with the EventNexus system are:

- User sign-up, secure (JWT) authentication and OAuth 2.0
- User roles: Admin, Organizer, Participant
- Organisations and events creation, scheduling and management
- Event recommendation using AI and predictive attendance modeling
- Integration of secure online payment processor and QR code ticketing
- Event reporting and analysis dashboard - this will help Program Managers to see how successful the events they are running will be.
- Chatbot for Attendee Support
- Multitenant system with tenant isolation.

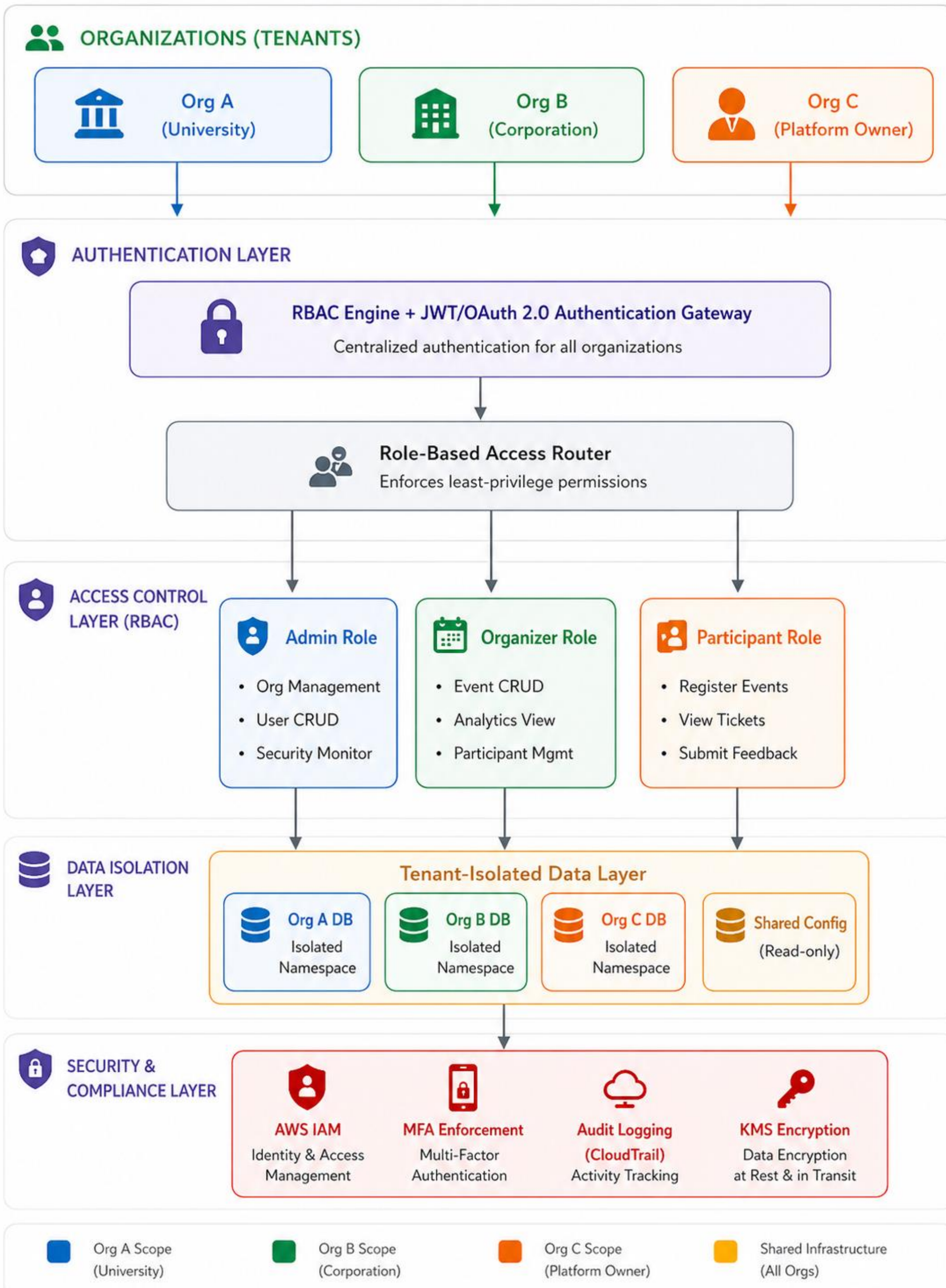
Nature of Challenges

Some of the design challenges of the EventNexus project are:

- Design of a secure multi-tenant system on the cloud and complete data segregation between the organisations without affecting the performance of the system
- Training and running the AI algorithm for predicting the attendance and recommendation of events
- Design of realizability for RBAC/IAM policies to different administrative structures of the companies
- Efficient use of cloud resources, particularly for server-less and container components, while making sure there is a fine balance between cost and scalability with a reasonable amount of resources.
- Providing a suitable degree of user experience for the users with different levels of technical proficiency

Figure 1:

EventNexus Multi-Tenant Architecture and RBAC Security Model



Overview of This Investigation Report

The investigation report is divided into the following four chapters:

- Chapter 1 - Introduction to the research study, covering background, problem background, project goal, and objectives; scope, potential benefits, target users, deliverables, and project plan.
- Chapter 2 - Literature Review on domain research into the application of AI in event management, cloud-based systems, multi-tenant security architectures, modelled systems, and technically justifying the choice in technology stack.
- Chapter 3 - Methodology covers the development approach of hybrid Waterfall–Prototyping methodology, design of the data gathering like survey instrument, and an analysis framework for the user requirements.
- Chapter 4 - Conclusion elaborates on what have been the outcomes from the investigation, the gaps found, and further areas for exploration in the development phase.

Project Plan

The Project EventNexus will be carried out through a series of structured phases, each having defined milestones and deliverables. The development process will begin with requirements gathering and design of system architecture, move through iterative development of the prototypes, functional implementation, security hardening, testing, deployment over AWS, and final documentation. Details of the timeline pertaining to all the phases are included as part of the appendices to this report in a Gantt chart.

Chapter 2: Literature Review

Introduction

This chapter critically evaluates prevailing literature in competitive research and systems on the implementation of EventNexus, an AI-Enabled Secure Cloud-Based Event Management System in multi-organization collaboration. The literature review of this chapter will be divided into three principal parts: domain research, evaluation of comparable systems, and alternative approaches to currently available systems, in which the emphasis will be put on the strengths and weaknesses of such systems.

Domain Research

Artificial Intelligence in Event Management

Some of today's event management's most game-changing technologies are AI. Thus, disruptive indeed, AI changes the very core of event organization, delivery, and evaluation in a contemporary fashion. AI has become an inevitable technology in modern event management. It helps boost planning, management, and evaluation processes for the event. Research has claimed that the role of AI through predictive analysis in chatbots, intelligent communication, and real-time monitoring in ensuring the improvement of efficiency with regards to the totality of the activity involved in an event lifecycle is very significant.

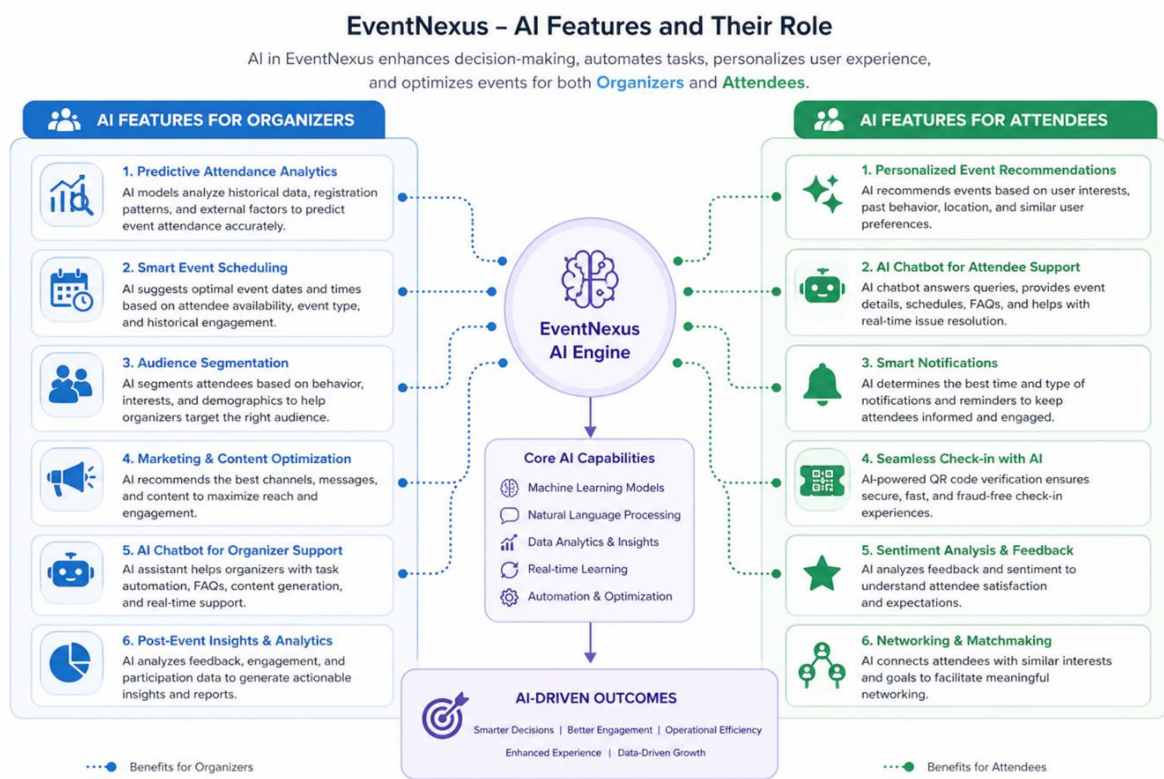
One of the most critical applications of AI is predicting attendance. It combines past event data and user behavior to incentivize or estimate whom and how many people will be attending. This helps the organizers plan their resources for an event, such as the venue, staff, and materials necessary, better (AndyM, 2026). This approach reduces guesswork and enables better decision-making.

The other very important concept is about the event-recommendation system. In this case, the systems recommend the events and urge users to participate based on their preferences and past activities just to raise their own engagement and satisfaction levels. Actually, studies show that this increases the attendance rate by two times and makes the users' experience better than with any other strategies (Marketing - Science topic, 2026).

There are also AI chatbots which usually help answer basic questions regarding the event details, enrollment, or schedule, hence relieving the organizers of pressure and offering feedback within a short period. Reports from the industry confirm that chatbots can automate most customer-support tasks and in general enhance operational efficiency (Mackenzie, 2026). The biggest impact on how immensely improved the process of event management could be seen, particularly when dealing with administrative overhead, was the reduction of huge amounts of time.

Figure 2*:*

EventNexus AI Features and Their Roles for Organizers and Attendees



The adoption of AI, globally, at the market level is also seen to be further showcasing the momentum rolling in for this technology to be integrated into events. According to a recent detailed market analysis, the value of AI in event management was USD 1.8 billion back in 2023, with a forecasted compound annual growth rate of about 22.9% until 2033, in which it will hit USD 14.2 billion. Although 69.1% of the market share was based on cloud deployments in 2023, it further underscores the continued robust growth of AI with the rise in cloud infrastructure (market.us, n.d.).

Cloud-Based Event Management Systems

Cloud computing is the lifeblood of the modern event technology landscape. It is a vibrant, easily adjustable, and cost-effective mechanism that integrates seamlessly with virtual and hybrid events alike. The most significant advantage of cloud-based execution is the ability to scale computing resources in and out as per requirement.

Event management, on the other hand, is like a rollercoaster in terms of traffic; it shoots up during registration and hits near zero after events conclude. The fine granularity in resource allocation and pay-per-use billing for consumed resources makes event-driven applications hugely cost-effective - especially when they have to scale up so much more power to respond to incoming loads of events.

The acreage of abstracting infrastructure management from development teams, it gives more room to develop the logic of an application than in capacity planning, which is done at the platform level based on the demand signal in real-time.

Further research on serverless computing across major cloud providers, namely AWS, Microsoft Azure, and Google Cloud Platform, research verified that these platforms are going to save great costs for an application that has irregular and unpredictable patterns of use. In an event management platform, that will be a real saving in operational cost amount since there are one thousand registrations at a quite small window and then mostly nothing-clearly saved operational costs in terms of total amount compared with the saved-server-type infrastructure.

Consequently, cloud platforms help in real-time data access and collaboration across geographically distributed teams and organizations. Many users from different organizations can work on shared event data concurrently with access control policies, but not with the same challenges of synchronization speed when compared to on-premises solutions.

Multi-Tenant Architecture and Cloud Security

It is Software-as-a-Service model deployment where many customers or tenants use a single instance of an application and its infrastructure with logical separation for their respective data and configurations. Such architecture has been adopted not only to save the costs involved and make management of the application easier but also in quest to ensure more security for the system.

The multi-tenant architectures mostly get employed in cloud-based systems, where numerous numbers of organizations gain access to one application at a time. This reduces cost and increases system efficiencies.

Security, however, is a very critical aspect when dealing with such multi-tenant environments. In the absence of good controls, one organization can actually gain access to another's data. This is achieved through systems that apply some form of data isolation techniques in ensuring each organization can only access its data. One such effectual approach to security is the role-based access control, where based on roles such as Admin, Organizer, Participant, etc., some permissions can easily be provided so unauthorized entry into the system is limited. This is an effective technique to enhance system security (Sharma, 2024).

In terms of Identity and Access Management (IAM), best practice in the industry dictates that to maintain security in a multi-tenant environment, multi-factor authentication (MFA), the least privilege principle, and audit logging of user actions, access requests, and security incidents within the context of all tenants should be implemented (Porter, 2024). These measures make up the core security controls of the EventNexus architecture.

Although there are advanced techniques in place, such as the possibility of applying machine learning to anomaly detection, this is far beyond the scope of our project. But indeed, this is where the future improvements can jump off of (Li et al., 2025). While beyond the scope of this project, this research gives valuable insight into future security improvements of EventNexus.

Similar Systems/Works

Eventbrite

Eventbrite is one of the most popular cloud-based event management systems in the world, offering event planners services for creating, marketing, registering and charging for events. Its easy-to-use interface and worldwide reach have established it as a market leader in the consumer events sector. But Eventbrite's features are mainly geared towards ticketing and attendee registration. The system has few AI capabilities - it does provide analytics, but advanced AI-based features such as predictive modelling of event attendance, participant recommendations, or the use of AI chatbots are not included. Moreover, Eventbrite does not support enterprise-level multi-organizational collaboration, and lacks the multi-tenancy architecture and role-based access control (RBAC) needed to facilitate secure data sharing across multiple organisations.

Cvent

Cvent is an enterprise event solution that is targeted at large organisations with complex event portfolios. It offers powerful event promotion, registration management, venue management, and event reporting capabilities. Cvent's analytics tools are more feature-rich than Eventbrite's and provide some data-based insights. But its AI capabilities are nascent - it does not offer machine learning and predictive analytics "first class" features such as machine learning-driven recommendations. Additionally, the Cvent architecture is costly to set up and manage, and is out of reach for small-to-medium enterprises and universities. Its support for multi-organizations is also weak in terms of granular, tenant-level data management.

Whova

Whova is a mobile-centric event management app designed to boost engagement and networking at face-to-face and hybrid events. It provides event agenda, announcement, networking and survey features. Whova offers a good user experience and has successfully been adopted in the academic conference market. But it has no significant cross-organizational collaboration capabilities and its security framework is not rich enough for an enterprise RBAC-based security model. Its AI capabilities are minimal and are limited to basic matchmaking, with no predictive analytics or machine learning.

Comparative Summary of Similar Systems

Below is a comparison of the three systems against the design principles of EventNexus:

Table 1:

Comparative Summary of Similar Systems

Feature	Eventbrite	Cvent	Whova	EventNexus (Proposed)
Cloud-Based Architecture	✓	✓	✓	✓
AI-Powered Analytics	Limited	Limited	Partial	✓ Full Integration
Multi-Organization Support	Limited	Partial	None	✓ Full Multi-Tenancy
RBAC / IAM Security	Basic	Moderate	Basic	✓ Advanced RBAC+IAM
Predictive Attendance	None	None	None	✓ ML-Based
AI Chatbot Support	None	None	None	✓ Integrated
QR Digital Ticketing	✓	✓	Limited	✓ Secure QR
Serverless Scalability	Partial	Partial	Limited	✓ AWS Serverless
Open-Source / Customizable	No	No	No	✓ Fully Custom

The review indicates that while each of the platforms considered address some of the needs for event management platforms, none integrate AI intelligence, multi-tenant security, and scalable cloud computing in a single platform. This confirms the need for EventNexus to be developed as a bespoke platform that integrates all these features.

Technical Research

This describes the technical research undertaken to identify, assess and justify the best combination of technologies used to build EventNexus. This selection was based on the project's requirements, including scalability, security, AI integration, real-time processing and maintenance. These factors are considered for each technology, drawing from academic and industry sources.

Programming Language and Framework Selection

JavaScript and the MERN Stack

EventNexus is going to be developed using a programming language JavaScript and the backend framework Node.js and Express.js and the frontend library React.js. This technological stack - often called the MERN stack (MongoDB, Express.js, React.js, Node.js) - is a popular full-stack web application development architecture.

This combination of technologies makes it possible to create scalable, secure and efficient web applications, which are tested by a study on the use of the MERN stack to develop e-commerce platforms (Patil et al., 2025). The ubiquity of JavaScript on the frontend and on the backend layers erases cognitive switching costs to developers, increases the productivity and reuse of developers on the application stack.

The event domain is also well fitted to the Node.js with its non-blocking event-driven input/output architecture that allows a server to maintain thousands of simultaneous connections without starting a new thread to handle each connection. This is necessary when there are spikes in event registration where the system may be able to handle thousands of simultaneous users. This is corroborated by the works of MDPI in the form of the article that demonstrates that the applications developed using Node.js can support high-concurrent tasks and deliver consistent response times.

The component-based approach of React.js enables the development of reusable user interface (UI) components that can be updated dynamically without having to reload the page. The outcome is an interactive user interface that will improve user experience. Vite and

ReactJS have been found to enhance the interactivity and speed of applications, which can be served by a server with a high capacity to serve large populations with a consistent user experience (Kandalkar et al., 2023).

Database Management System: MongoDB

The database management system that will be used by EventNexus system will be MongoDB. MongoDB is a document-oriented, NoSQL database with flexible, JSON-like BSON documents. This schema free character fits best in an event management system where objects like events, bookings and users can change in their properties and properties may have to be added or dropped as the system continues to grow.

With its horizontal scalability (particularly the sharding option) built-in, MongoDB enables the database infrastructure to expand to multiple servers, which will ensure that the system will be capable of supporting the increasing number of users without being affected by the performance. The article by MDPI Engineering Proceedings, which assesses the MERN stack as a collaborative web app, confirms that the scalable NoSQL technology of MongoDB provides an excellent base to multiple content storage and retrieval processes and that it is a good fit when the data volume in the system is large (Sravani et al., 2024).

The fact that Node.js and MongoDB can be easily integrated through the Mongoose Object Document Mapper (ODM) also makes development easier since data schema and validation constraints can be defined and implemented in JavaScript and therefore the stack is more consistent in the language used.

Cloud Platform: Amazon Web Services (AWS)

Cloud computing forms a key part of the modern event management systems, which in turn provides a very flexible and scalable base for any application. By hosting applications online, it gives users access to them from virtually anywhere in the world, namely seamlessly, hybrid, and so on.

EventNexus will be deployed on the Amazon Web Services (AWS), which is the market leader in cloud services. The decision to use AWS was informed by the fact that it offers a wide spectrum of services, is reliable and considers security and scalability in the context of the proposed architecture.

One of the benefits of cloud systems is scalability. Most often, event platforms might experience a peak in traffic at the time of registration and may have very minimal use after that. Of the two, that first one can move through that smoothly by the enablement of resources to be scaled up or down based on demand, hence at event time becoming cost-effective (Borra & Pamidipoola, 2025).

Other services provided by AWS that offer solutions to the security requirements of EventNexus include AWS IAM to provide fine-grained access control. The cloud system also subscribes to real-time data access and collaboration. Different users from different organizations can access and manage the event's information at the same time. This enhances coordination and conversation compared to conventional systems.

In general, capability, availability, and flexibility are better under the clouds than in traditional on-premises systems, making them very appropriate for modern event management platforms.

Authentication and Security: JWT and OAuth 2.0

The multi-tenant model in EventNexus will also need a strong user authentication and authorisation. EventNexus will be based on the security protocols JSON Web Tokens (JWT) and OAuth 2.0.

In a paper published in the MDPI journal Computers about enhancing JWT authentication, it is revealed that JWT - specified in RFC 7519 - is a stateless and compact means of transmitting and communicating authentication claims through a secure method (Bucko et al., 2023). JWT being stateless means that the server does not need to maintain user state, which adds to the scalability of a system that has many users concurrently. It can also be conveniently integrated with AWS services and microservices by the use of tokens.

Another study article on the application of OAuth2 and JWT in distributed API gateways confirms that OAuth 2.0 provides a secure method of authorization by isolating the user authentication process and app authentication process to enable third-party services access the EventNexus APIs without requiring user authentication. This plays an important role in the payment gateway and external notification services. JWT and OAuth 2.0 provide a multi-layered methodology of authentication and authorisation in security and scalability.

AI and Machine Learning Tools

EventNexus AI skills will be deployed through a mixture of TensorFlow.js (in-browser and model inference with Node.js) and Python-based machine learning packages such as Scikit-learn to train and assess models. The full machine learning ecosystem of Python (NumPy, Pandas, Matplotlib, etc.) is mature enough to use it as a machine learning language of choice in data preprocessing, model training, and performance analysis.

Three main AI capabilities will be powered by these tools: an attendance prediction model, which will be trained on historical registration and attendance data; a collaborative filtering recommendation system, which will suggest events in terms of participant preference profiles; and a rule-based and ML-hybrid chatbot, which will answer frequent questions of participants. As a holistic industry survey confirms, the three categories of AI application have been demonstrated to be the most important in terms of performance enhancement throughout the event management landscape.

Development Tools

The following development tools have been selected to support the development process:

- Visual Studio Code - the primary integrated development environment, which has extensive JavaScript/Node.js support, debugging, and Git integration.
- Git and GitHub - for version control, collaborative development, and code review workflows
- Postman - to test and document API in a RESTful manner.
- Docker - to containerize application components in the development and testing of the applications to maintain consistency in the environment.
- AWS Management Console and CLI - to provision, configure and monitor cloud.

Technical Research Summary

Justifications of the choice of the MERN technology stack, MongoDB, AWS cloud infrastructure, JWT/OAuth 2.0 authentication, and TensorFlow.js/Python AI tools have been thoroughly justified in this chapter. All the components were chosen based on the academic evidence, as well as the industry best practice, which makes sure that EventNexus is developed on the basis of the technically sound and scalable, yet security-conscious base. The chosen technologies are both compatible with each other and fit together as a set suited to the requirements of a complex, AI-enabled, multi-tenant cloud application.

Summary

This chapter has laid down a very strict theoretical and comparative base in the development of EventNexus. The reviewed literature confirms that AI technologies, such as predictive analytics, recommendation systems, and chatbots, have proven to offer relatively high levels of benefits to the efficiency of event management and the experiences of participants. Cloud computing provides the scalability of infrastructure and cost flexibility needed by modern events and multi-tenant architecture patterns with RBAC and IAM solutions meet the security and data isolation demands of multi-organizational settings.

Chapter 3: Methodology

System Development Methodology

One of the very critical decisions within the software project lifecycle is the selection of the best system development methodology for requirements elicitation, system design, development, and quality assurance. This chapter talks about the methodology that was picked for the development of EventNexus and makes a strong academic justification for that choice, and then further details its practical application at each of the stages of development.

Methodology Selected: Hybrid Waterfall–Prototyping

It is a hybrid methodology that mixes the Waterfall model with iterative Prototyping and was chosen for the development of EventNexus. The reason for this choice was that neither methodology was good enough to embrace all the complexity of this project: the Waterfall model was good at a solid structure of rigor and discipline in documentation that serves the base for any multi-component enterprise system, and Prototyping renders iterative development cycles user-centric in a way effective for validating AI features and usability across a diverse set of users.

The research completed for the Sustainability journal concludes that it is possible to overcome some limitations of each separate methodology through the combination of structured phase-gate progression, as in Waterfall, with agile elements, such as Rapid Prototyping. As recognized by the research, the stakeholder alignment, and the completeness of documentation between given stages, assured by formal sign-off requirements of the Waterfall model, empowered the team to embody its ideas along with feedback from users before moving them to the next step. It greatly reduces the possibility of very expensive changes during later stages (Senarath, 2021).

There is very good straight evidence to go for that choice: hybrid methodologies were noted by the publication to be so fitting, most suitably so, for projects where clear upfront planning is needed at the same time as there is a need to adapt - which is exactly the profile of EventNexus. It has some well-specified architectural requirements, favoring the clarity given by Waterfall, but also some AI features and user experience design elements that would benefit prominently from iterative user feedback, hence favoring Prototyping.

Justification of the Selected Methodology

Structured Development with Clear Phase Boundaries

The Waterfall hybrid method very well makes sure that after each development phase - requirements analysis, system design, implementation, testing, deployment - it is taken to a given standard before the next phase is initiated. Indeed, research in hybrid software development methods confirmed that his sequential structure brings improved predictability to the project and hence in following progress, managing budgeting, and retaining documentation comprehensively.

This then means that formal verification checkpoints, inherently contained in Waterfall, are needed for quality assurance in a project so complicated like EventNexus; after all, it deals with security-related aspects such as IAM and multi-tenant isolation.

User-Centered Validation Through Prototyping

The Prototyping part overcomes a second major drawback of pure Waterfall development: every single user experience decision made cannot be validated in advance before full implementation. The prototype is developed extremely fast either as a functional mock-up or a working system prototype and exposed to representative users for evaluation, the feedback of which is fed back into successive iterations before it is finalized for development (Maharao, 2022).

User rejection of the final system. It follows that prototyping will be most effective in reducing the ambiguity between requirements and in minimizing the risks associated with user rejection of the final system. This, therefore, becomes important for EventNexus especially in the areas of the AI recommendation interface, event dashboard design, or participant registration flow-where intuitive usability would count toward adoption success.

Risk Mitigation

This will help the team pinpoint technical and usability issues early on-when they are cheaper to fix, since the approach used is to prototype these high-risk or uncertain features early in their development before full implementation. It is most valuable for the AI modules: meaningful evaluation of model performance can only really be done with the right data and user interaction.

Enhanced Stakeholder Involvement

This is strongly supported by the prototyping stages that are already built into almost any development effort. It allows direct evaluation of the system prototype, thereby ensuring formal correctness of the technically correct system to be delivered and that it is fully used, which proved to be one of the most frequent reasons for default in projects developed by a strictly sequential approach without internal validation of users.

Balance Between Flexibility and Structural Control

Flexibility, Structural Control The research identifies that hybrid agile-structured methodologies work most effectively once a project has leveled off - when it plateaus to a steady state, giving the opportunity for further features to be prototyped within a plan-based schedule. This is much the kind of development environment in which EventNexus finds itself; it is well acquainted with core architectural components, such as database schema, cloud infrastructure, and authentication, but needs predictability in Waterfall for AI features and User Experience prototyping.

Development Phases

Phase 1: Requirements Analysis

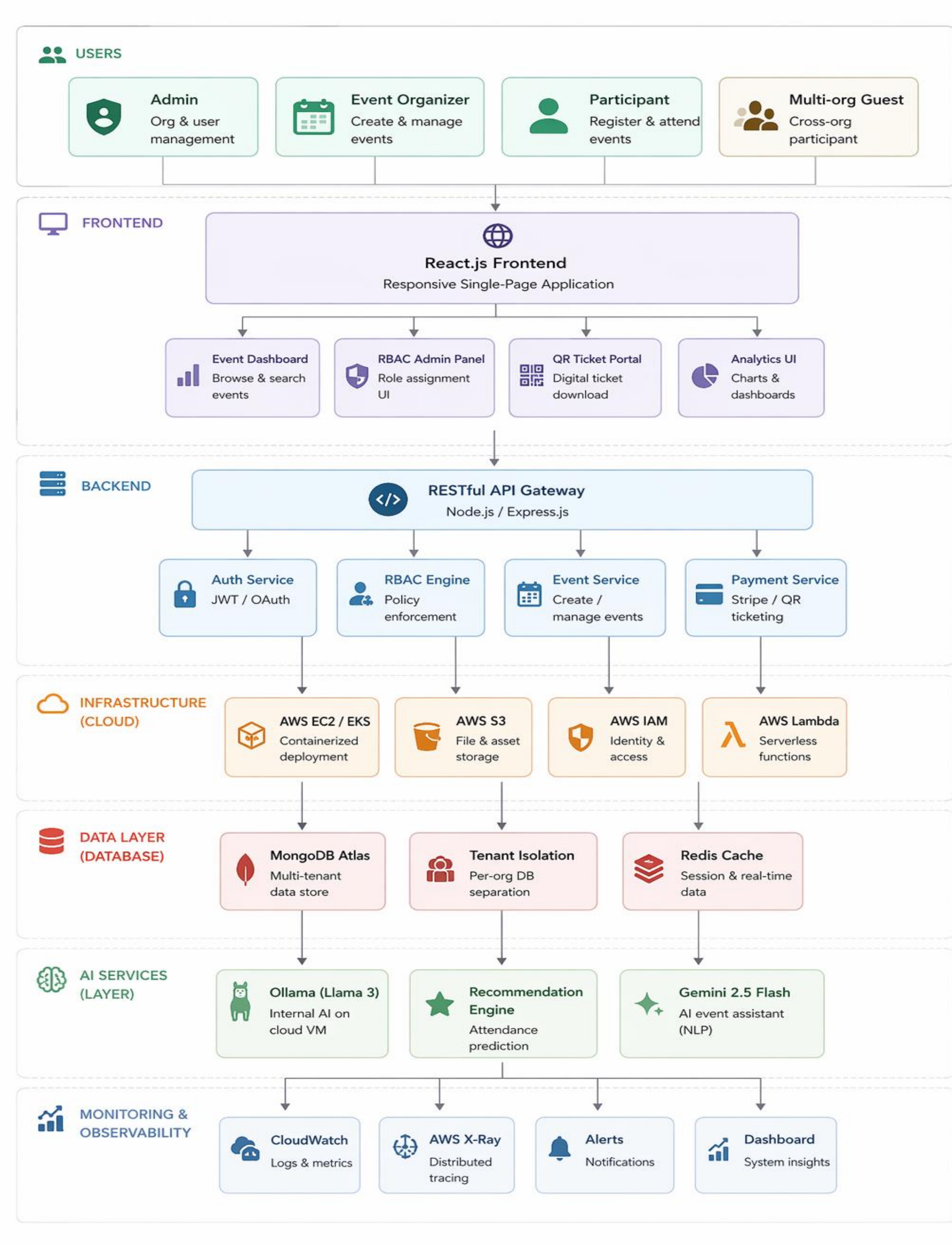
System requirements are elicited from all key stakeholders through structured questionnaires and interviews. Comparable systems will be analyzed with an aim to gather requirements for both system functionality and non-functionality. There should be the validation of requirements toward project objectives before moving to phase two.

Phase 2: System Architecture and Design

It is the detailed explanation of the system architecture. It covers the multitenant database schema, cloud infrastructure topology, RBAC permission matrix, specification of API endpoints, and wireframes of the UI. The security design - JWT/OAuth 2.0 authentication flow and tenant isolation mechanisms - shall be defined during this phase and then shall be reviewed against predefined norms or frameworks.

Figure 3:

EventNexus Overall System Architecture



Phase 3: Prototype Development and User Evaluation

At this point, all key features-for example, the event management dashboard, AI recommendation interface, and participant registration flow-are prototyped at a functional level and opened up to the users for validation. The feedback is recorded, processed, and further iterations of adaptations are made to the prototype plus used to refine the system specifications before the full design process is initiated. This phase might take another two or three cycles of iteration until the prototype gains acceptance.

Phase 4: Implementation

Development on the full system is done after the detailed design of Phases 2 and 3 to the final specifications of the system. The development follows an incremental approach; first of all, the authentication and RBAC infrastructure should be developed, and core functionalities for event management developed on top of that will include AI features and payment/ticketing subsystems. Each of the modules is checked through the unit tests before integration.

Phase 5: Integration Testing and Quality Assurance

Extensive testing is performed in three different areas: functional testing assures that all described functions are working properly, security testing guarantees correct enforcement of RBAC, proper authentication, and intact tenant isolation, whereas performance testing assures that the system will meet all scalability requirements even at peak loads simulated during stress testing. The defects are captured and corrected long before the model goes into production.

Phase 6: Deployment and Monitoring

The system is implemented into the AWS cloud platform in the form of containerized elements (Docker) and serverless functions (AWS Lambda) accordingly. AWS CloudWatch is used to set up real-time performance monitoring and alerting. User acceptance testing is carried out after deployment with a sample of administrators, organizers and participants to ensure that the system works properly in the production environment.

Summary

This chapter has given a detailed explanation as to why the hybrid Waterfall-Prototyping development methodology was chosen to be used in EventNexus. The approach is validated through academic literature on ResearchGate, MDPI, and ShodhKosh to make the best-suited solution to addressing the distinctive demands and risk profile of the project. One is the structural rigor and documentation discipline to buttress security-critical enterprise systems, and the other is user-focused iterative validation that makes it possible to support AI functionality and user experience design. The methodology was translated into a feasible, executable project roadmap during the six-phase development plan.

Data Collection Method

System requirements for EventNexus were validated through a quantitative survey approach based on primary data collection. It would give data in structuring experiences, preferences, and expectations of potential users on an event management system.

A total of seventy-two responses came from students, event attendees, event organizers, and administrators. Accordingly, the results identified the existing pain points in the current systems of events and helped clarify which features were considered most important by users.

Survey Design

An online questionnaire was designed, with close-ended questions that participants would answer on a rating scale or choose multiple-choice options. The survey questions ranged from the frequency of event participation to satisfaction with the existing systems, problems during registration, features required in a given platform, security needs, A.I based recommendations, cloud data storage, notifications, ticketing systems using QR code, and inter-organizational collaboration.

Justification of Data Gathering Technique

A survey was considered relevant for the purpose of data collection as this method of data collection is efficient for the large user group. In-app survey questions are often quantified so they can be directly related to interviews or observations in such a way that it will help the overall statistical analysis process. For validation of system requirements within EventNexus, this approach was particularly useful in measurement of user needs, feature preferences, and expectations at scale.

Survey Analysis

Most of the respondents are active people in general, with 61.1% of such people attending or organizing events "Often" or "Very Often." The greatest number among them, 48.6%, were Students/Participants, followed by the Event Organisers at 30.6% and the Administrators at 16.7%.

The present levels of satisfaction with the existing event management platform can be considered moderate, as 37.5% of them are ranked at level 3, while merely 6.9% get the full mark of 5. The most commonly mentioned issues are problems with badly organized events, lack of reminders, and technical issues, each at 33.3%; thereafter comes the challenge of a registration process, composed of about 25%, wherein security is at 19.4%.

Secure login and cloud-based storage lead the list of most necessary features, with 95.8% being considered "Important" and "Very Important" to the participants. Event reminders and event notifications came next at second place: 95.8% assigned the same rating-that these were required.

QR code ticketing had a "Very Useful" or "Useful" usage of 87.5%, above, while AI and machine learning-based events picked up very high acceptance with an 86% score for being useful.Online registration emerged as the most popular preferred feature, at 31.9%, followed by AI Recommendations at 26.4%, Secure Login at 25%, and Multi-Organisation Collaboration at 23.6%.

Moreover, 83.3% would "Definitely" or "Probably" use an AI-driven and cloud-based event management solution. Not one of them ticked "Definitely No." We, therefore, require a more intelligent, integrated event solution by EventNexus.

Chapter 4: Conclusion

The investigation has laid down a holistic academic and technical groundwork for an AI-Enabled Secure Cloud-Based Event Management System for Multi-Organization Collaboration. The literature review in Chapter 2 has identified and justified the major issues in existing event management systems, verified the suitability of AI and cloud computing in addressing these issues, and established the security need of multi-tenant systems. The technical research chapter also validated the chosen technology stack, the MERN stack, AWS cloud computing, MongoDB database, JWT/OAuth 2.0 authentication and TensorFlow.js, as suitable and proven technologies for developing the system.

The methodology chapter justified the Waterfall-Prototyping development method and developed the project development plan in six phases. The data collection design provided the survey tool and validated it as appropriate to gather primary data on user requirements from the target user group. Overall, these chapters have shown that the study has been conducted sufficiently and with the required academic rigour to form the foundation for the next phase of this project.The student plans to refine and build on some aspects in the subsequent phase. The AI components, especially the attendance predictor and the collaborative filtering recommendation system cannot be fully tested, as they currently do not have access to historical event data for training and validation.

Likewise, the multi-tenant security model although justified theoretically, will need to undergo empirical security testing and penetration testing during the implementation phase to ensure the proposed RBAC and IAM security controls work as planned. This presents areas for further research in the development and experimentation stages of the project.

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Appendix A – PPF

Figure 4:

Cover Page of PPF

Student Name: Anjali Mishra

Supervisor: Er. Shambhu Gautam

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Program: BSc.IT

Title: AI-Enabled Secure Cloud-Based Event Management System for Multi-Organization Collaboration

Figure 5:

Introduction (PPF)

AI-Enabled Secure Cloud-Based Event Management System for Multi-Organization Collaboration

Introduction

Cloud computing is a way of delivering computing services - such as servers, storage, databases, networking, and software - over the internet on demand, allowing users to access resources quickly without managing physical infrastructure. This model allows scalability, flexibility, and saving cost, and is therefore suitable to systems which are expected to serve a huge number of users or fluctuate as workload varies.

Over the past years, artificial intelligence (AI) has become more and more a part of cloud systems to automate their tasks, improve the process of decision-making, and discover patterns within data that can be useful. The AI can assist the systems to learn with the activity of users and enhance scheduling, communication, and reporting tasks. These features render AI a useful complement to the cloud-based applications that deal with hard workflows and extensive data volumes.

The issue of security in cloud computing environment presents certain challenges due to the fact that third parties host data and applications and access to the internet. These are configuration risks, unauthorized access as well as identity and access management requirements to secure sensitive information. Cloud security is a set of activities and technologies that are aimed at protecting the data, systems, and networks on the cloud.

Artificial intelligence enhances cloud security by enabling real-time monitoring, anomaly detection, and adaptive responses to potential threats. These capabilities are important for cloud-based systems operating in shared environments, where user activity and access patterns frequently change. In multi-organization event management platforms, AI-enabled security helps ensure system reliability, data protection, and secure collaboration during high-traffic events.

Planning and conducting of events like conferences, training, and seminars in which those participating and the organizers may be in different teams or institutions is one of the many organizations that must collaborate on the planning and execution of the same.

Figure 6:

Problem statement (PPF)

Conventional event management systems might fail to provide secure multi-organizing collaboration, smart automation, which results in the necessity of a cloud-based system with the integration of AI-based capabilities and robust security standards and the multi-organization community of users.

Problem Statement

With an increasing number of systems going online, it is important to guarantee the security of data and services stored in the cloud. Cloud environments are by nature shared and have open access on the internet which subjects systems to vulnerability to security risks like misconfigurations, unauthorized access and breach of data. Securing cloud resources therefore needs a well-organized concept that involves access controls, monitors and secure setups.

Traditional rule-based security methods are often ineffective in dynamic, multi-tenant cloud environments. Rapid changes in user behavior during event planning and execution make it difficult to detect security threats using static controls. Without AI-driven monitoring and automated response, security incidents may go undetected, increasing the risk of unauthorized access and data breaches.

Multi-tenant architecture (where a single application instance is used by many clients) is required when several organizations are using the same event management platform. In this type of systems, the privacy and protection of individual organizations (tenants) should rest with them, even though they are stored on the same platform. Breach of data may occur or improper access between organizations due to failure in isolation of tenants or control of access (Multitenancy, 2025).

Multi-organization systems are also more complicated in identity and access management (IAM) since users differ in their roles and access privileges. The IAM solutions should be such that only authorized users are allowed to access the relevant resources and unauthorized users are not allowed to access those. Cloud identity services are used to offer central authentication and control of access and this assists in enforcing policies among users and applications (Microsoft Entra ID , 2026).

Figure 7:

Aim, Objectives and Literature Review (PPF)

Conventional event management platforms mostly rely on manual operation in executing activities such as scheduling, communication and reporting, thus, inefficiencies and delays. Smart automation - e.g., AI-based notifications, analytics, refinements of tasks, etc. - is required to enhance the responsiveness of systems and minimize the human resources to coordinate events with multiple organizations (Common Cloud Computing Terminology, 2023).

Aim

To develop a secure, AI-enabled cloud-based event management system that supports collaborative event planning and execution across multiple organizations.

Objectives

- To design and develop a secure cloud-based event management platform that supports collaboration among multiple organizations.
- To implement role-based access control (RBAC) and identity and access management (IAM) to ensure secure and controlled access for different user roles.
- To integrate AI-enabled features for automation and analytics, providing intelligent insights to support event planning and performance evaluation.
- To utilize cloud-native and serverless technologies to achieve scalability, reliability, and cost efficiency under varying workloads.
- To provide a centralized dashboard for real-time monitoring of events, users, and system activities.

Literature Review

Cloud computing can be given as the provision of computing services through the internet which is scalable, flexible and can be accessed remotely without running physical infrastructure. This model allows users to access resources on demand and only pay means used making it appropriate in scalable multi-user applications such as event management.

Figure 8:

Literature Review (PPF)

Clouds with multiple tenants enable one instance of an application to be used by a number of consumers known as tenants. Although this option makes infrastructure less expensive and easier to manage, it comes with the difficulty of data isolation and access control, as shared environments have to guarantee that the data of each of the tenants is secure and distinct (Multitenancy, 2025).

Cloud security is a set of policies, technology and practices used to secure information and infrastructure on the cloud. Encryption, access control and monitoring are security controls that can be used to protect cloud systems against threats. Another factor in cloud computing security is a shared responsibility model in which the provider secures core infrastructure as the customers handle their data and access control (Cloud computing security, n.d.).

The control of identities and access to cloud environments is necessary to regulate access to system resources. The cloud-based IAM solutions are more centralized in their authentication and authorization and, therefore, the importance of user privileges and protection of sensitive information in multi-organization structures are easier to manage (Microsoft Entra ID , 2026).

The latest research on the AI-driven event management websites focuses on how cloud-based applications, role-based access control (RBAC), and smart automation could be utilized to improve safety and teamwork. Serverless computing enhances the scalability and reliability when events are ongoing, whereas AI-based functionalities like event insights, automated content creation, and smart proposals enhance the effect of decision-making and being present (Honey John, 2024).

Recent studies indicate that combining artificial intelligence with cloud security improves threat detection, proactive risk management, and automated incident response. Machine learning-based anomaly detection and predictive analytics are particularly effective in securing multi-tenant cloud applications. These approaches support continuous monitoring, data isolation, and secure collaboration in cloud-based systems.

Figure 9:

Deliverables (PPF)

Deliverables

Cloud-Hosted Event Management Application

- Web application that is fully operable on a cloud platform.
- Multi-organizational scalable architecture.

AI-Powered Features

- Automated notifications (e.g., reminders, updates)
- Basic analytics (attendance stats, participation trends)

Secure Identity and Access Management (IAM)

- Unique roles (admin, organizer, participant) of the user login system.
- Employing a centralized access control over data.

Multi-Organization Collaboration Tools

- Scheduling and management tools of shared events.
- Individual organization data view in order to ensure privacy.

Central Dashboard

- Account of events, users and activities.
- Management and organizer visual reporting.

Figure 10:

References (PPF)

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Appendix B – PSF

Figure 11:

PSF

STUDENT ID : NP069810

INTAKE ID : NP3F2509IT(CE)

STUDENT NAME : Anjali Mishra

A. Project Title

"EventNexus: AI-Enabled Secure Cloud-Based Event Management System for Multi-Organization Collaboration"

B. Brief Description of Project Background

Problem Context

Digital technologies have been embraced fast in the event management industry due to the need to keep up with the globalization and hybrid events, as well as the popularity of cloud systems.

Majority of the inter-organizational partnerships lead to the creation of conferences, academic seminars, corporate workshops, and training programs.

However, the remnants of the traditional event management system can still be seen in the issues of having to do manual coordination of events, not having intelligent automated features, low security, and limited scalability.

Moreover, papers on AI-based event platforms discuss how planning accuracy, attendee engagement, and the operational efficiency of the event can be improved to a great extent through such platforms (John et al., 2024).

Besides that, researches on the application of AI in event planning revealed technologies like predictive analytics, automated scheduling, recommendation systems, and chatbot support features to help enrich the participant's experience (Abdul & Abdul, 2023).

Cloud-based event management systems are scalable and can be accessed from anywhere, a much-needed feature that primarily helps virtual and hybrid events (Event et al., 2019). However, multi-organizational collaboration brings with it some new challenges such as:

- Data must be separated among tenants
- Identity and Access Management (IAM) complexity
- Role-based permission enforcement
- Real-time security monitoring
- Secure payment integration

Multi-tenant cloud architecture effectively helps to reduce the operating costs; however, without proper configurations, it even increases the chances of unauthorized access and data leakage (Cheerla & Chandrasena, 2024).

Studies on the integration of role-based access control within cloud security environments indicate that defining access privileges based on user roles significantly reduces unauthorized access and minimizes data leakage, thereby strengthening an organization's overall security posture (Jude & Titi, 2025).

Other than that, recent ERP-cloud integration research mainly targets business cost reduction and operational efficiency improvement when AI is deeply integrated into cloud business systems (John & Isaac, 2025).

Therefore, there is a clear need to develop:

- A secure
- AI-enabled
- Cloud-native
- Multi-organization event management platform

that integrates automation, intelligent analytics, strong RBAC security, and scalable cloud infrastructure (Kumar & Ritesh).

Rationale

According to the identified challenges, the proposed system EventNexus expects to combine the Artificial Intelligence and secure cloud computing to deliver: Intelligent event planning support (AI recommendations, attendance prediction)

- Smart event planning support (recommendations, predictions of attendance)
- RBAC and IAM Secure multi-tenant architecture.
- Scalability on large events of cloud.
- Live Analytics Dashboards.
- Assured payment and ticket validation with QR.

Studies indicate that AI-based event systems improve decision-making and minimize the efforts of administrators with predictive analytics and automation. Event systems deployed on clouds enhance flexibility, cost-effectiveness and accessibility remotely (Maurya et al., 2025).

Smart event portals powered by AI with recommendation systems add personalization and satisfaction to the user.

Moreover, the research in the fields of cloud ERP and AI integration shows that AI lowers the operational expenses and resource distribution in clouds (Halim & Abdul, 2023).

EventNexus is therefore an advancement of the traditional event systems to a digital collaboration ecosystem that is intelligent, secure and scalable.

Tangible Benefits

- Reduction in operational and administrative costs
- Improved event attendance prediction accuracy
- Secure digital ticketing and automated verification
- Reduced manual scheduling and coordination workload
- Scalability during peak event registration periods

Intangible Benefits

- Improved user satisfaction through AI recommendations
- Enhanced trust due to strong security architecture
- Better collaboration between organizations
- Data-driven decision-making
- Modern, innovative digital event experience

Nature of Challenge

Developing EventNexus involves several technical and organizational challenges:

To begin with, it is hard to design a safe multi-tenant cloud architecture and, at the same time guarantee complete isolation of data between organizations.

Second, the implementation of AI modules (recommendation systems, predictive analytics, and chatbots) presupposes the appropriate preparation of the dataset and the choice of algorithms.

Third, the adoption of RBAC and IAM in several organizations should be properly designed in terms of policy.

Fourth, to maintain cloud scalability without raising operational cost, it is vital that serverless architecture or container-based architecture is used optimally.

Lastly, the challenge of ensuring usability of the system without sacrificing the level of advanced AI is another important issue.

C. Brief description of project objectives.

Deliverables

EventNexus: AI-Enabled Secure Cloud-Based Event Management System is a collaboration between multiple organizations to plan, manage and analyze events within a centralized secure cloud-based platform. The system incorporates the capabilities of the Artificial Intelligence (AI) with the cloud computing to improve automation, scalability, and decision-making in the event management processes.

Through this application, the administrators, event organizers, and participants will have access to the system safely depending on the roles assigned to them. With the help of AI-based analytics, organizations will be able to create and manage events, track registrations, and analyze performance. The participants will be equipped to explore events, get personal suggestions, and book the events through integrated authentication systems and payment systems with utmost security.

The following are the main functions of the system that will be realized prior to delivery.

Admin

- o Enable the administration to create and administer several organizations in the system.
- o Enable user to add and modify user accounts.
- o Enable an administrator to assign roles (Admin, Organizer, Participant) through Role-Based Access Control (RBAC).
- o Enable the administrator to view the system activity logs and security events.
- o Enable option to see analytics reports by AI modules created by the administrator.
- o Enable the administrator to implement tenant-level data isolation to enable a secure multi-organization collaboration.
- o Enable the administration to manage event approvals and system setups.

Event Organizers

- o Permit the organization of events, updating as well as management.
- o Enable organizers to plan event sessions and the event details.
- o Enable the organizers to handle the registration of participants.
- o Provide AI attendance prediction reports to organizers.
- o Enable organizers to use automated messages and reminders to participants.

o Only organization-specific data can be accessed by the organizers.

Participants

o Enable participants to create and log in using authentication protocols (JWT/OAuth).

o Enable the participants to search through the events.

o Enable members to get AI-based suggestions of personal events.

o Enable members to pay online by way of tickets to secure events.

o Enable participants to install digital tickets in the form of QR.

o Permit the participants to provide feedback on the completion of the event.

AI Features

o Prediction of attendance using previous data pattern.

o Event popularity analysis and participation analytics.

o Smart suggestion system among the participants.

o AI chatbot response to event-related queries and make the application interesting.

D. Brief Description of the Resources Needed by the Proposal

(Hardware, Software, Access to Information/Expertise, User Involvement)

Hardware

The minimum requirements for hardware to successfully develop and implement the EventNexus system are as follows:

- Processor – Intel Core i3 or higher
- Random Access Memory (RAM) – Minimum 4GB (8GB recommended)
- Storage – 256GB SSD
- Keyboard & Mouse
- Stable Internet Connection (RJ45 / Wireless Fidelity – Wi-Fi)

Given that the system will be hosted on a cloud platform, there will be no need to have heavy infrastructure in the area. The cloud service provider will take care of most storage, scaling, and processing requirements.

Software

The minimum software requirements for the development and execution of the project are as follows:

Database Management System (DBMS)

- MongoDB

Server-Side and Application Development

- Programming Language – JavaScript
- Backend Framework – Node.js with Express.js
- Frontend Framework – React.js
- API Communication – RESTful APIs
- Authentication – JSON Web Token (JWT) / OAuth

Cloud Platform

- Amazon Web Services (AWS)

Development Tools

- Code Editor – Visual Studio Code
- Version Control – Git

Documentation and Planning

- Microsoft Project
- Microsoft Word

AI and Analytics Tools

- TensorFlow.js
- Chart.js / Plotly.js
- Python

Access to Information

Experts who have experience in cloud computing, integration of artificial intelligence, and development of secure web applications will be consulted. The necessary information in the project will be obtained in the form of academic literature review, documentation of the cloud provider (AWS/Azure), and research articles concerning the AI-powered event systems, cloud security, and multi-tenant architecture.

User Involvement

The system will be designed based on the needs of various interests such as the administrators, event organizers, and participants representing various organizations. Potential users will be interviewed on the prototyping and testing stages to get feedback. They will be taken into account to enhance their usability, security and general performance of the system. The

\

involvement of users will be highly crucial especially at the stage of usability testing and evaluation.

E. Academic Research Being Carried Out and Other Information, and Techniques Being Learned

To carry out the deliverables, the preliminary list of books and online resources that will be studied are as follows.

Books

Name: *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow.*

Author: Aurélien Géron

Publisher: O'Reilly Media

Name: *Learning React: Modern Patterns for Developing React Apps*

Authors: Alex Banks and Eve Porcello

Publisher: O'Reilly Media

Name: *Web Application Security: Exploitation and Countermeasures for Modern Web* **Applications**

Author: Andrew Hoffman

Publisher: O'Reilly Media

Name: *Practical Machine Learning with Python: A Problem-Solver's Guide to Building Real-World Intelligent Systems.*

Authors: Dipanjan Sarkar, Raghav Bali, and Tushar Sharma

Publisher: Apress (Springer Nature)

Online Resources

- Honey John (2024), "AI Powered Event Management Platform", IRJMETS.
- "Cloud-Based Event Management: Organizing and Hosting Virtual Events", ResearchGate, 2023.
- "AI Driven Smart Engineering Event Portal with Multi-Agent Recommendation and Analytics", CSEIT, 2025.
- "Cloud-Based ERP and AI Integration for Cost Optimization", ResearchGate, 2024.
- Hussain & Khan (2022), "AI and Cloud Security Synergies: Building Resilient Information and Network Security Ecosystems", ResearchGate.

F. Brief Description of the Development Plan

System Development Methodology

A system development methodology can be regarded as a set of what would essentially be very formalized step-by-step guidelines on which a set of people in an organization adheres to when they make the decision to arrive at an entirely new information system through the formation of a completely new system. To develop EventNexus, we will use a hybrid approach between the Waterfall and the Prototyping methods.

The traditional approach, waterfall, will assist us in passing through the development process more disciplined, in a highly sequenced manner. The phases of development are so sequenced such that the successful completion of one stage requires the commencement of the other and phases do not overlap. Thus, the project will conclusively pass through phases which are numerically and clearly-defined like requirement analysis, system design, implementation, testing and deployment. In that way, there is no need to say that it is highly easy to monitor the level of project development and adequate documentation will become much easier and less tense.

On the same note, the Prototyping technique will enable us to develop a working prototype of the system interfaces and AI functionality with simply going through a rapid development process, and very early testing. Consequently, we can identify any potential problems, receive free user feedback, and virtually remove the development risk.

Therefore, by combining these two methodologies, the entire development lifecycle will be a mixture of predictability and agility.

G. Brief Description of the Evaluation and Test Plan

Success Criteria

The primary objective of the article EventNexus: AI-Enabled Secure Cloud-Based Event Management System is to offer secure multi-organization event management, along with AI-powered analytics and role-based access control.

- It will be said that the system is successful in case: Users can securely register and login according to assigned roles.
- Organizations can manage events independently without data leakage.
- AI-based predictions and recommendations function correctly.
- The system handles concurrent users efficiently.
- The dashboard generates accurate analytics reports.

To ensure the system is functioning correctly before delivery, users representing different roles will participate in testing as follows:

Unit Testing

In this type of testing, each individual module of the system will be tested separately to ensure it meets the specified requirements. Modules such as user authentication, event creation, AI analytics, and payment integration will be tested independently.

For example, in the login module, the system will verify correct credential validation, token generation, and role assignment before allowing access.

Each unit must pass the testing phase before being integrated into the complete system.

Integration Testing

During integration testing, various unit-tested modules will be connected and tested as a whole in order to ensure that the front-end, back-end, database, and AI (Analytical Intelligence) components are functioning well together.

For example, looking from an organizer's standpoint, after the creation of an event, it is necessary that the event data gets correctly saved in the database and then can be seen on the participant's dashboard.

Usability Testing

Artificial intelligence also needs to be able to provide accurate prediction results that should be displayed on the analytics dashboard. They will be assessed through the GUI, the navigation structure, the response time and the feedback message (Usability, 2023).

The system should be designed in a human-centered way so that even the average user would be able to comprehend it without getting professional assistance. The feedback gathered during this phase will serve to enhance the whole user experience and make it more fluid.

Appendix C – Ethics Form

Figure 12:

Ethics Form

Office Record	Receipt – Fast-Track Ethical Approval
Date Received:	Student name: Anjali Mishra Student number: NP069810
Received by whom:	Received by: Date:

APU / APIIT FAST-TRACK ETHICAL APPROVAL FORM (STUDENTS)

Tick one box (level of study):

- ☐ POSTGRADUATE (PhD / MPhil / Masters)
- ☐ UNDERGRADUATE (Bachelors degree)
- ☐ FOUNDATION / DIPLOMA / Other categories

Tick one box (purpose of approval):

- ☐ Thesis / Dissertation / FYP project
- ☐ Module assignment
- ☐ Other: ____________

Title of Programme on which enrolled … …B.Sc. IT (CE)… … …

Tick one box: ☐ Full-Time Study or ☐ Part-Time Study

Title of project / assignment AI Enabled Secure Cloud-Based Event Management System For multi-organization collaboration.

Name of student researcher … …Anjali… Mishra… … …

Name of supervisor / lecturer… Mr. Shombhu Gautam … … …

Student Researchers- please note that certain professional organisations have ethical guidelines that you may need to consult when completing this form.

Supervisors/Module Lecturers - please seek guidance from the Chair of the APU Research Ethics Committee if you are uncertain about any ethical issue arising from this application.

		YES	NO	N/A
1	Will you describe the main procedures to participants in advance, so that they are informed about what to expect?	✓		
2	Will you tell participants that their participation is voluntary?	✓		
3	Will you obtain written consent for participation?	✓		
4	If the research is observational, will you ask participants for their consent to being observed?	✓		
5	Will you tell participants that they may withdraw from the research at any time and for any reason?	✓		
6	With questionnaires and interviews will you give participants the option of omitting questions they do not want to answer?	✓		
7	Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?	✓		
8	Will you give participants the opportunity to be debriefed i.e. to find out more about the study and its results?	✓		

If you have ticked **No** to any of Q1-8 you should complete the full Ethics Approval Form.

		YES	NO	N/A
9	Will your project/assignment deliberately mislead participants in any way?		✓	
10	Is there any realistic risk of any participants experiencing either physical or psychological distress or discomfort?		✓	
11	Is the nature of the research such that contentious or sensitive issues might be involved?		✓	

If you have ticked **Yes** to 9, 10 or 11 you should complete the full Ethics Approval Form. In relation to question 10 this should include details of what you will tell participants to do if they should experience any problems (e.g. who they can contact for help). You may also need to consider risk assessment issues.

			YES	NO	N/A
12	Does your project/assignment involve work with animals?			✓	
13	Do participants fall into any of the following special groups? **Note that you may also need to obtain satisfactory clearance from the relevant authorities**	Children (under 18 years of age) People with communication or learning difficulties Patients People in custody People who could be regarded as vulnerable People engaged in illegal activities (eg drug taking)		✓	
14	Does the project/assignment involve external funding or external collaboration where the funding body or external collaborative partner requires the University to provide evidence that the project/assignment had been subject to ethical scrutiny?			✓	

If you have ticked **Yes** to 12, 13 or 14 you should complete the full Ethics Approval Form. There is an obligation on student and supervisor to bring to the attention of the APU Research Ethics Committee any issues with ethical implications not clearly covered by the above checklist.

STUDENT RESEARCHER

Provide in the boxes below (plus any other appended details) information required in support of your application, THEN SIGN THE FORM.

Please Tick Boxes

I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU Research Ethics Committee.	✓
Give a brief description of participants and procedure (methods, tests used etc) in up to 150 words. EventNexus is a secure, AI enabled cloud system that helps multiple organizations planand manage users and view AI analytics, organizers create and schedule events, and participants register & get AI based recommendations. The system is developed using waterfall & Prototyping methods, with testing for usability, functionality & security. AI tools supports attendance prediction & analytics. By combining cloud scalability & AI Intelligence as well as role-based access, EventNexus makes event management easier, more secure & efficient for all organization involved.	
I also confirm that: i) All key documents e.g. consent form, information sheet, questionnaire/interview are appended to this application.	✓
Or ii) Any key documents e.g. consent form, information sheet, questionnaire/interview schedules which need to be finalised following initial investigations will be submitted for approval by the project/assignment supervisor/module lecturer before they are used in primary data collection.	✓

E-signature... ... Anjali ... ... Print Name... Anjali Mishra ... ... Date..24th Feb, 2026
(Student Researcher)

Please note that any variation to that contained within this document that in any way affects ethical issues of the stated research requires the appending of new ethical details. New ethical consent may need to be sought.

The completed form (and any attachments) should be submitted for consideration by your Supervisor/Module Lecturer

SUPERVISOR/MODULE LECTURER
PLEASE CONFIRM THE FOLLOWING:

	Please Tick Box
I consider that this project/assignment has no significant ethical implications requiring a full ethics submission to the APU Research Ethics Committee	✓
i) I have checked and approved the key documents required for this proposal (e.g. consent form, information sheet, questionnaire, interview schedule) Or	✓
ii) I have checked and approved draft documents required for this proposal which provide a basis for the preliminary investigations which will inform the main research study. I have informed the student researcher that finalised and additional documents (e.g. consent form, information sheet, questionnaire, interview schedule) must be submitted for approval by me before they are used for primary data collection.	✓

SUPERVISOR AND SECOND ACADEMIC SIGNATORY

STATEMENT OF ETHICAL APPROVAL (please delete as appropriate)

1) **THIS PROJECT/ASSIGNMENT HAS BEEN CONSIDERED USING AGREED APU/SU PROCEDURES AND IS NOW APPROVED**

2) **THIS PROJECT/ASSIGNMENT HAS BEEN APPROVED IN PRINCIPLE AS INVOLVING NO SIGNIFICANT ETHICAL IMPLICATIONS, BUT FINAL APPROVAL FOR DATA COLLECTION IS SUBJECT TO THE SUBMISSION OF KEY DOCUMENTS FOR APPROVAL BY SUPERVISOR (see Appendix A)**

E-signature... Shambhu ... ... Print Name.. Mr. Shambhu Gautam ... ... Date... 24th Feb, 2026
(Supervisor/Lecturer)

E-signature... ... ... Print Name. Er. Ashutosh Karn ... ... Date.. 24th Feb, 2026
(Second Academic Signatory)

Office Record	Receipt – Appendix A (Fast-Track Ethics Form)
Date Received:	Student name:
	Student number:
Received by whom:	Received by:
	Date:

APPENDIX A
AUTHORISATION FOR USE OF KEY DOCUMENTS

Completion of Appendix A is required when for good reasons key documents are not available when a fast track application is approved by the supervisor/module lecturer and second academic signatory.

I have now checked and approved all the key documents associated with this proposal e.g. consent form, information sheet, questionnaire, interview schedule

Title of project/assignment... An AI-Enabled Secure Cloud-Based Event Management System for multiorganization collaboration.

Name of student researcher ... Anjali Mishra

Student ID: NP069810 Intake: ... NP3F2509IT (CE)

E-signature... Shambhu ... Print Name. Mr. Shambhu Gautam Date. 24th Feb, 2026
(Supervisor/Lecturer)

Appendix D – Meeting Logs

Figure 13:

Log Sheet - Meeting 1

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Project Log Sheet – Supervisory Session

Notes on use of the project log sheet

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisor session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's Name: Anjali Mishra   Date: 24th Feb, 2026   Meeting No. 1

Project Title: AI Enabled secure CloudBased EMS for Multi-org collaboration.   Intake: NP3F2509IT(CE)

Supervisor's Name: Mr. Shambhu Gautam   Signature: Shambhu

Items for discussion (noted by student before mandatory supervisory meeting):

1. Project Topic : AI Enabled secure cloud base EMS for multi-Org. collaboration.
2. Selected MERN stack : MongoDB, Express.js , React.js , Node.js .
3. About the AI integration/implementation in the system.
4. Need for research papers on AI, cloud systems & multi-organization platform.
5.

Record of discussion (noted by student during mandatory supervisory meeting):

1. Supervisor approved the project topic and stack.
2. MERN stack is suitable for front-end and back-end development.
3. Research should focus on AI- Integration , cloud multi-org. collaboration & security.
4. Guidance on finding research papers: journals, Keywords & casestudy.
5. Recommended to select a base/ reference paper as a foundation for project design.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. Collect 10-15 research papers on AI cloudbased event system.
2. Identify one base reference paper for guidance.
3. Take notes on AI , cloud , and security techniques
4. from research papers .
5.

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Figure 14:

Log Sheet - Meeting 2

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Project Log Sheet – Supervisory Session

Notes on use of the project log sheet

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project (SIX mandatory supervisory sessions)
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session
3. A log sheet is to be brought by the STUDENT to each supervisor session
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter A copy is retained by the student to be filed in the project file
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session
7. The log sheet is an important deliverable for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively

Student's Name: Anjali Mishra   Date: 15th march, 2026 Meeting No: 2

Project Title: AI Enabled Secure Cloud Based EMS for Multi organization collaboration   Intake: NP3F2509IT(CE)

Supervisor's Name: Mr. Shambhu Gautam   Signature: ...SHambHUU...........

Items for discussion (noted by student before mandatory supervisory meeting):

1. Review of system flow and overall architecture.
2. Evaluation of current research papers used in the proposal.
3. Identification of required base papers.
4. Planning next development steps.
5.

Record of discussion (noted by student during mandatory supervisory meeting):

1. Presented the 7 layer system flow & architecture, approved by supervisor
2. The supervisor reviewed the research paper & suggested to find stronger base paper.
3. The supervisor advised using recent paper (2022-2026) from IEEE etc
4. The supervisor approved the architecture & asked the student to update the papers and AI model explanations before the next meeting.
5.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. Identify stronger research paper.
2. clearly define the AI module used for the system features.
3. Get chapter 1 reviewed.
4.
5.

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Figure 15:

Log Sheet - Meeting 3

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Project Log Sheet – Supervisory Session

Notes on use of the project log sheet

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project (SIX mandatory supervisory sessions).
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisor session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's Name: Anjali Mishra Date: 5th April, 2026 Meeting No: 3

Project Title: AI Enabled Secure cloud Based EMS for Multi organization collaboration. Intake: NP3F2509IT(CE)

Supervisor's Name: Mr. Shambhu Gautam Signature: Shambhu

Items for discussion (noted by student before mandatory supervisory meeting):

1. AI model selection
2. chapter 1 progress
3. System modules overview
4. Updated research paper review
5.

Record of discussion (noted by student during mandatory supervisory meeting):

1. AI module discussed, more research suggested.
2. Advised to clearly explain model integration.
3. Supervisor suggested to keep better comparison in literature.
4. more research paper needed as well as research in Git.
5.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. clearly define AI models
2. Update architecture diagram.
3. Revise chapter 1
4. start chapter 2.
5.

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Figure 16:

Log Sheet - Meeting 4

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Project Log Sheet – Supervisory Session

Notes on use of the project log sheet

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project (SIX mandatory supervisory sessions)
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session.
3. A log sheet is to be brought by the STUDENT to each supervisor session.
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form.
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter. A copy is retained by the student to be filed in the project file.
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session.
7. The log sheet is an important deliverable for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively.

Student's Name: Anjali Mishra Date: 16th April, 2026 Meeting No: 4

Project Title: AI Enable Secure Cloud Based EMS for Multi-organization collaboration Intake: NP3F2509IT(CE)

Supervisor's Name: Er. Shambhu Gautam Signature: Shambhu

Items for discussion (noted by student before mandatory supervisory meeting):

1. final literature review structure and comparison.
2. selection of system development methodology.
3. Proposed data gathering techniques (survey.., etc)
4. Draft system architecture & workflow.
5. Chapter 2 completion progress.

Record of discussion (noted by student during mandatory supervisory meeting):

1. Literature review refined with clearer comparison of existing system.
2. Approved selected methodology.
3. Proposed data gathering methods.
4. Initiated system architecture and workflow diagram review.
5. Supervisor suggested improving explanation of AI module.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. finalize chapter 2 with proper comparison & references.
2. Design survey questions.
3. start writing chapter 3 (methodology section)
4. Also, chapter 4 completing the conclusion section as well.
5.

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Figure 17:

Log Sheet - Meeting 5

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Project Log Sheet – Supervisory Session

Notes on use of the project log sheet

1. This log sheet is designed for meetings of more than 15 minutes duration, of which there must be at minimum SIX(6) during the course of the project (SIX mandatory supervisory sessions)
2. The student should prepare for the supervisory sessions by deciding which question(s) he or she needs to ask the supervisor and what progress has been made (if any) since the last session, and noting these in the relevant sections of the form, effectively forming an agenda for the session
3. A log sheet is to be brought by the STUDENT to each supervisor session
4. The actions by the student (and, perhaps the supervisor), which should be carried out before the next session should be noted briefly in the relevant section of the form
5. The student should leave a copy (after the session) of the Project Log Sheet with the supervisor and to the administrator at the academic counter A copy is retained by the student to be filed in the project file
6. It is recommended that students bring along log sheets of previous meetings together with the project file during each supervisory session
7. The log sheet is an important deliverable for the project and an important record of a student's organization and learning experience. The student must hand in the log sheets as an appendix of the final year documentation, with sheets dated and numbered consecutively

Student's Name: Anjali Mishra Date: 28th April, 2026 Meeting No: 5th

Project Title: AI Enabled Secure Cloud Based EMS for Multi-Organization Collaboration Intake: NP3F2509IT(CE)

Supervisor's Name: Er. Shambhu Gautam Signature:

Items for discussion (noted by student before mandatory supervisory meeting):

1. Completion of all chapters (chapter-1-4)
2. Finalization of survey results in the data collection & analysis
3. Documentation of survey result in the report.
4. Review of system design architecture & workflow.
5. Preparation of final project documentation for submission.

Record of discussion (noted by student during mandatory supervisory meeting):

1. Confirmed all 4 chapters are completed
2. Survey was successfully conducted and results were properly analyzed & included in the document.
3. Supervisor reviewed the final report and suggested minor formatting & content improvement.
4. System architecture, workflow & proposed solution were approved.
5. Overall documentation was accepted.

Action List (to be attempted or completed by student by the next mandatory supervisory meeting):

1. Make final correction as per feedback.
2. Improve formatting, referencing & consistency of the document.
3. Prepare & organize the final document for submission.
4. Submit the completed project documentation.
5.

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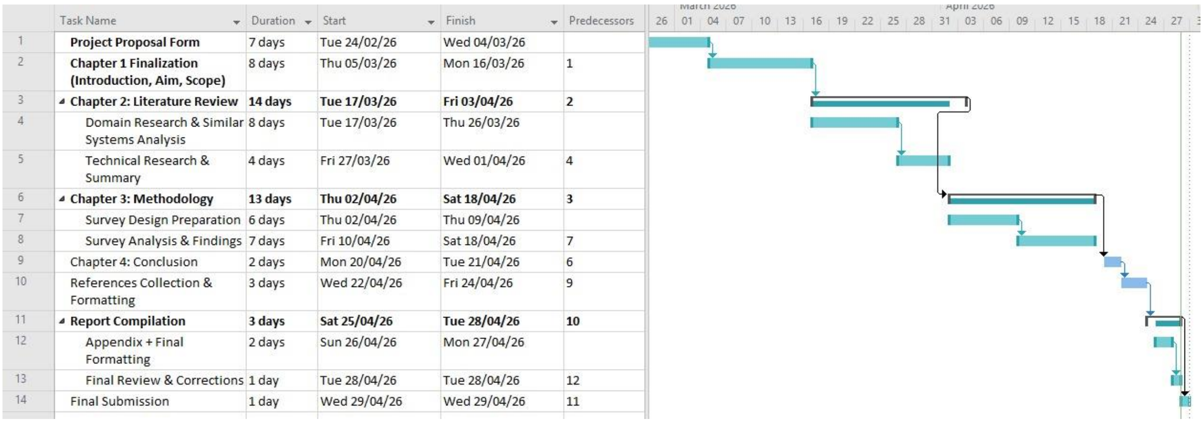
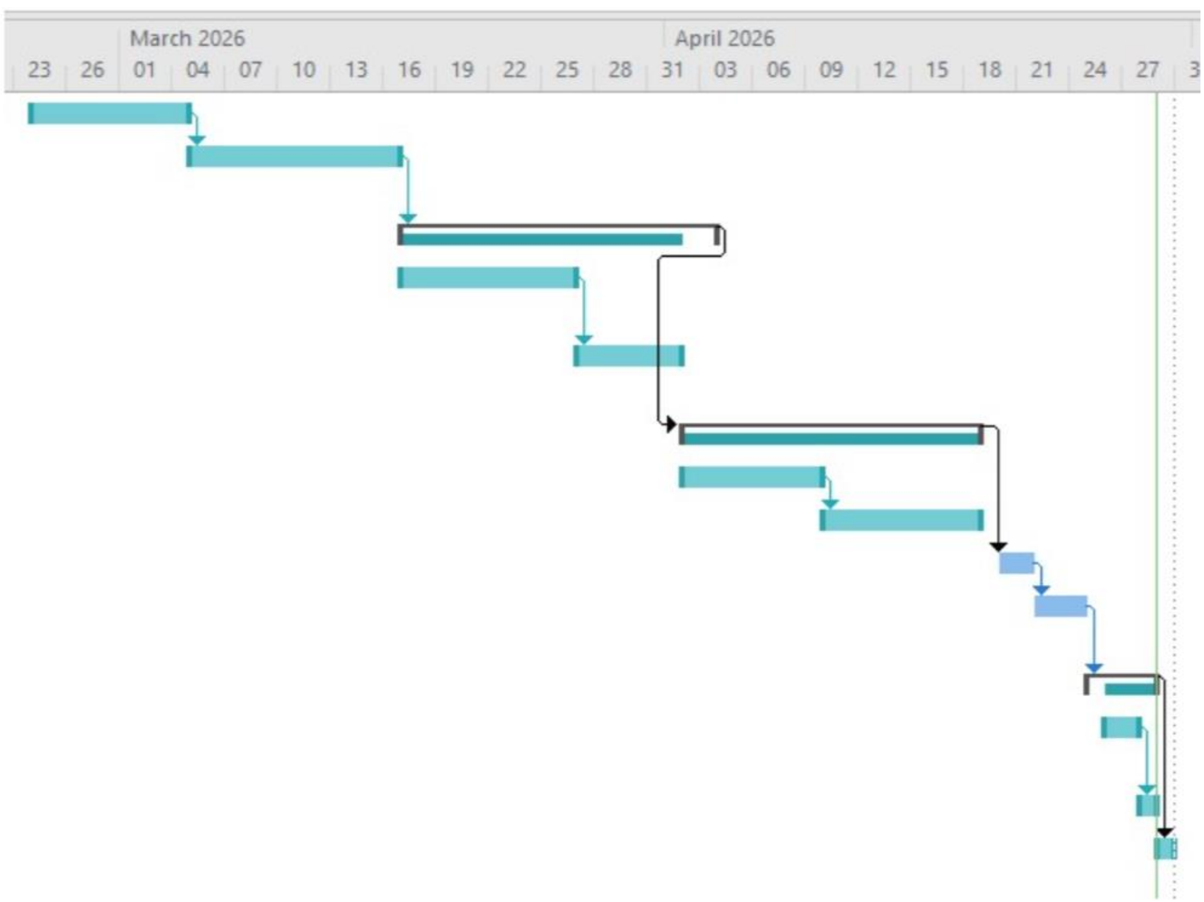
Appendix E – Gantt Chart

Figure 18:

Gantt Chart



	Task Name	Duration	Start	Finish	Predecessors
1	**Project Proposal Form**	7 days	Tue 24/02/26	Wed 04/03/26	
2	**Chapter 1 Finalization (Introduction, Aim, Scope)**	8 days	Thu 05/03/26	Mon 16/03/26	1
3	**Chapter 2: Literature Review**	**14 days**	**Tue 17/03/26**	**Fri 03/04/26**	**2**
4	Domain Research & Similar Systems Analysis	8 days	Tue 17/03/26	Thu 26/03/26	
5	Technical Research & Summary	4 days	Fri 27/03/26	Wed 01/04/26	4
6	**Chapter 3: Methodology**	**13 days**	**Thu 02/04/26**	**Sat 18/04/26**	**3**
7	Survey Design Preparation	6 days	Thu 02/04/26	Thu 09/04/26	
8	Survey Analysis & Findings	7 days	Fri 10/04/26	Sat 18/04/26	7
9	Chapter 4: Conclusion	2 days	Mon 20/04/26	Tue 21/04/26	6
10	References Collection & Formatting	3 days	Wed 22/04/26	Fri 24/04/26	9
11	**Report Compilation**	**3 days**	**Sat 25/04/26**	**Tue 28/04/26**	**10**
12	Appendix + Final Formatting	2 days	Sun 26/04/26	Mon 27/04/26	
13	Final Review & Corrections	1 day	Tue 28/04/26	Tue 28/04/26	12
14	Final Submission	1 day	Wed 29/04/26	Wed 29/04/26	11



Appendix F - Survey Questions and Responses

Following are the questionnaire and the survey results:

Figure 19:

How often do you attend or organize events?

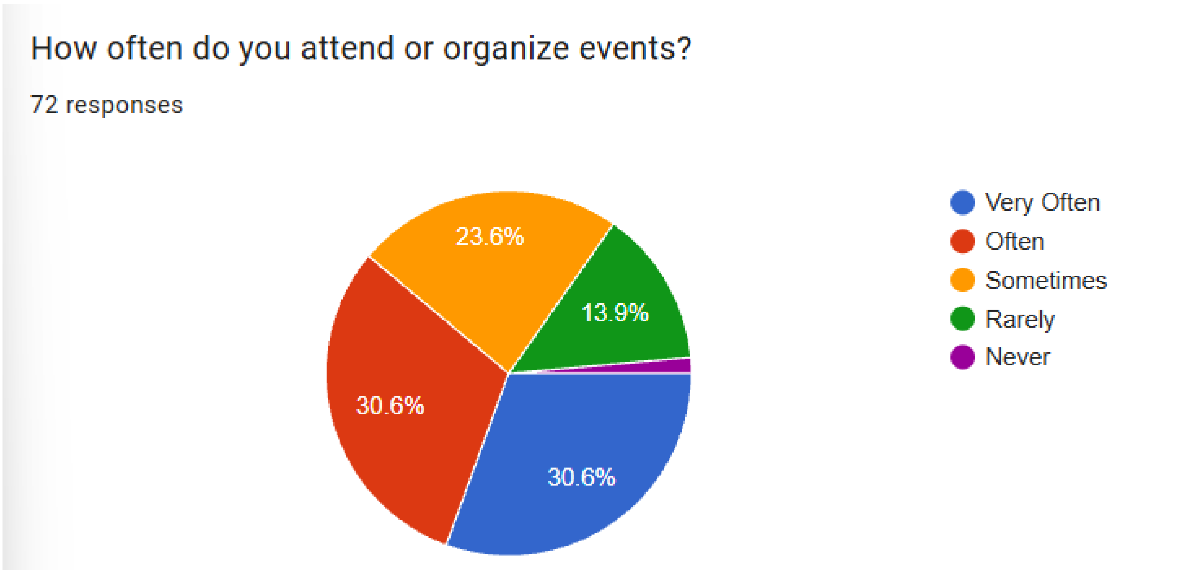


Figure 20:

What is your role in events?

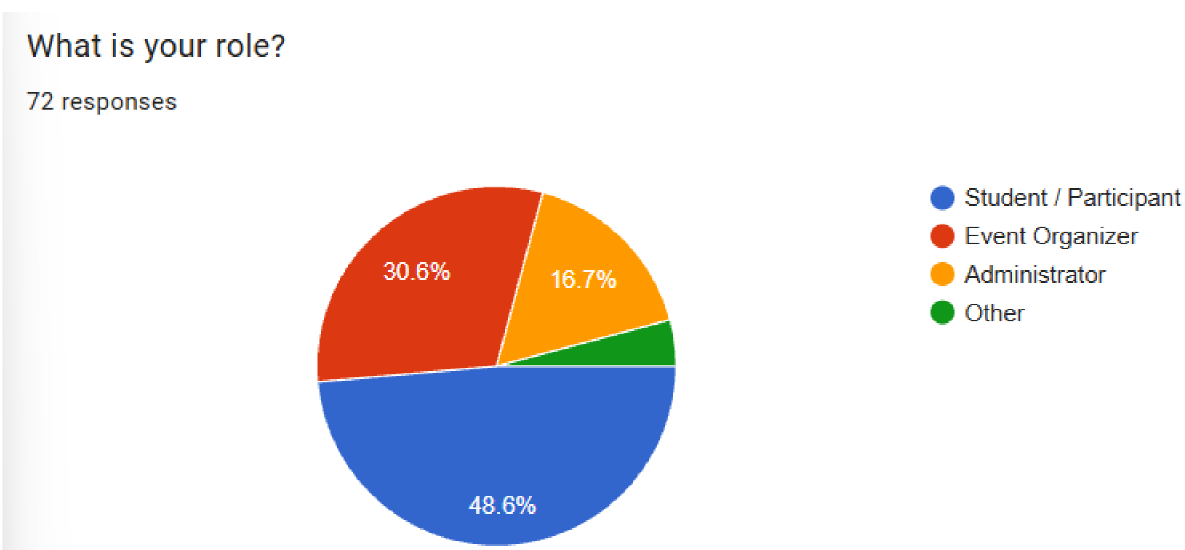


Figure 21:

How satisfied are you with current event management systems?

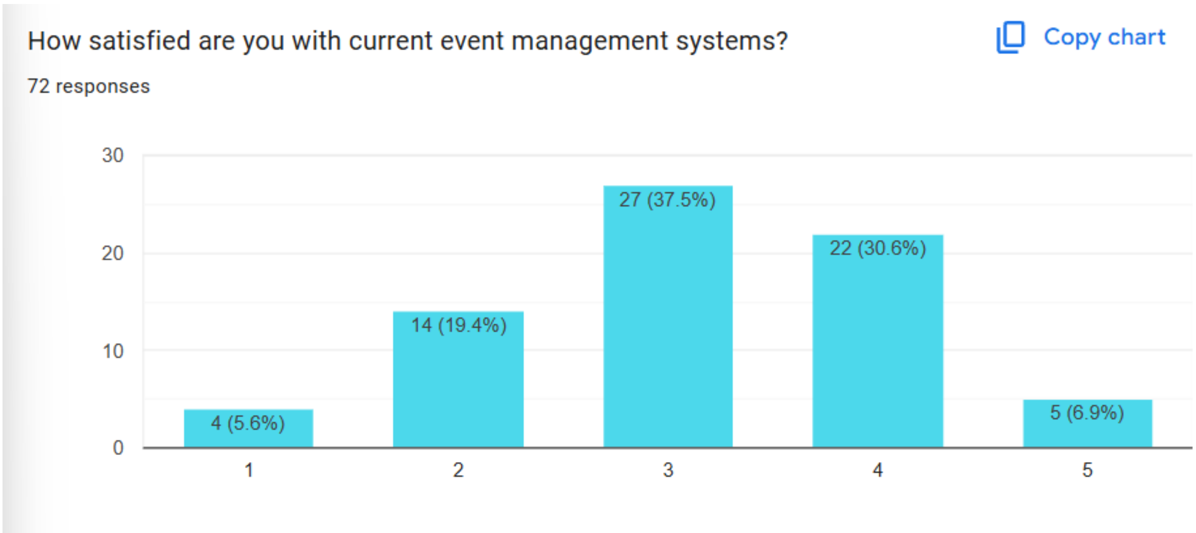


Figure 22:

How often do you face problems when registering for events?

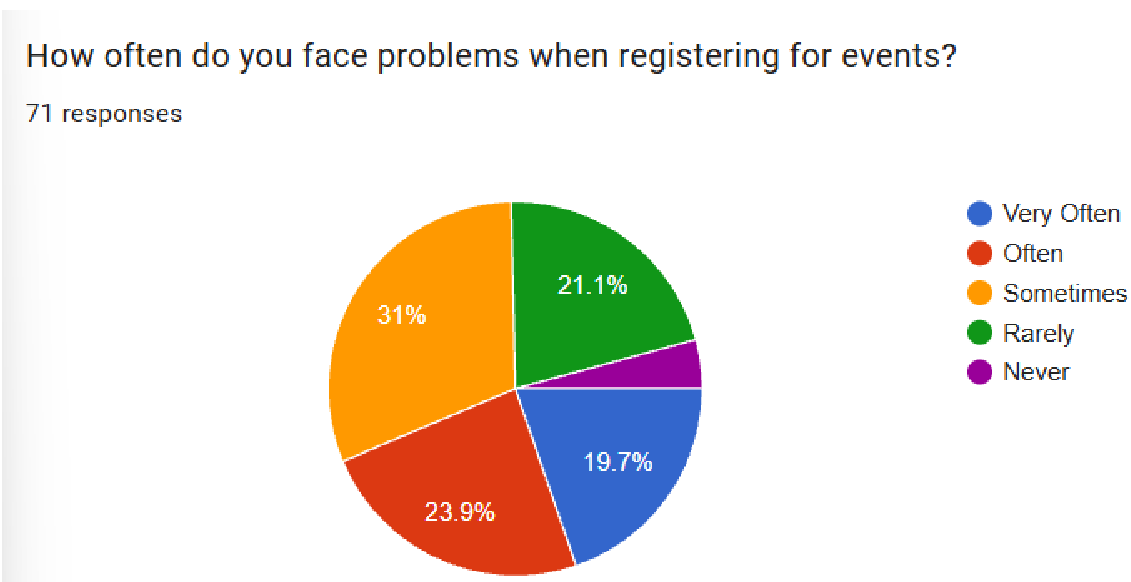


Figure 23:

Which problem do you face most in current event systems?

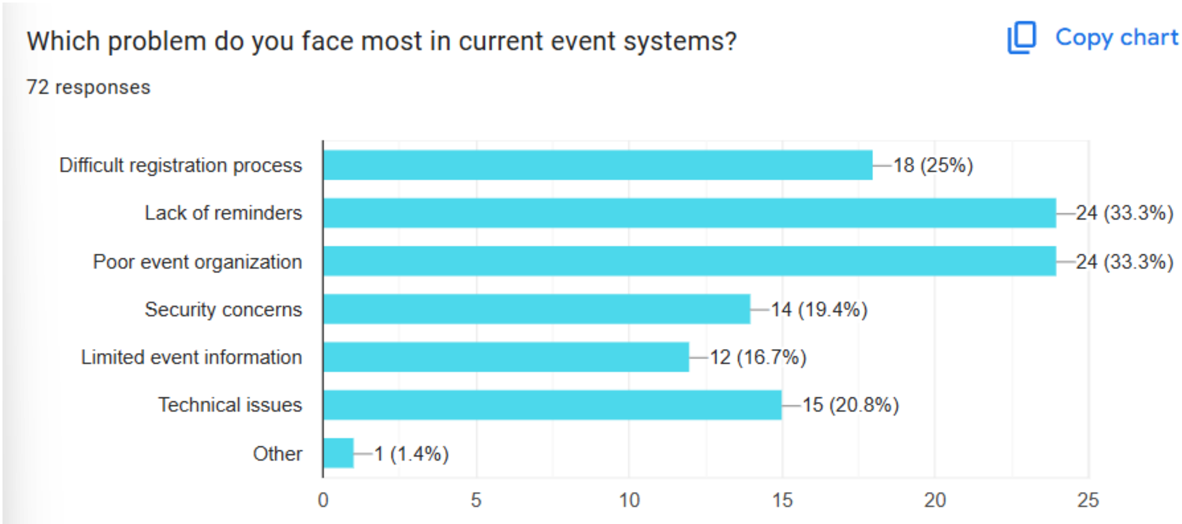


Figure 24:

How useful would shared event management between organizations be?

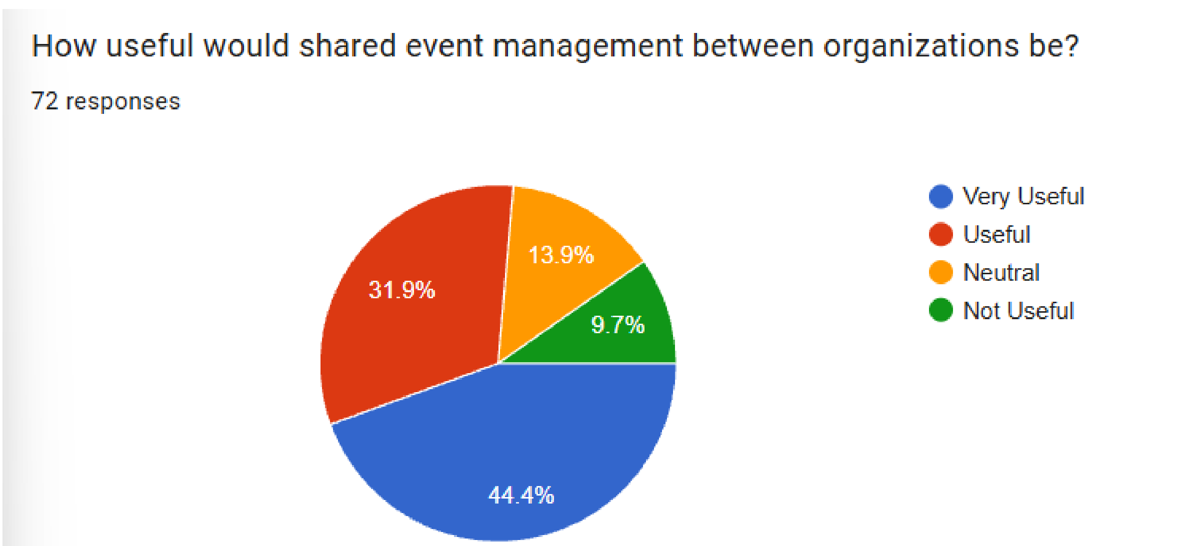


Figure 25:

How important is secure login and cloud-based data storage?

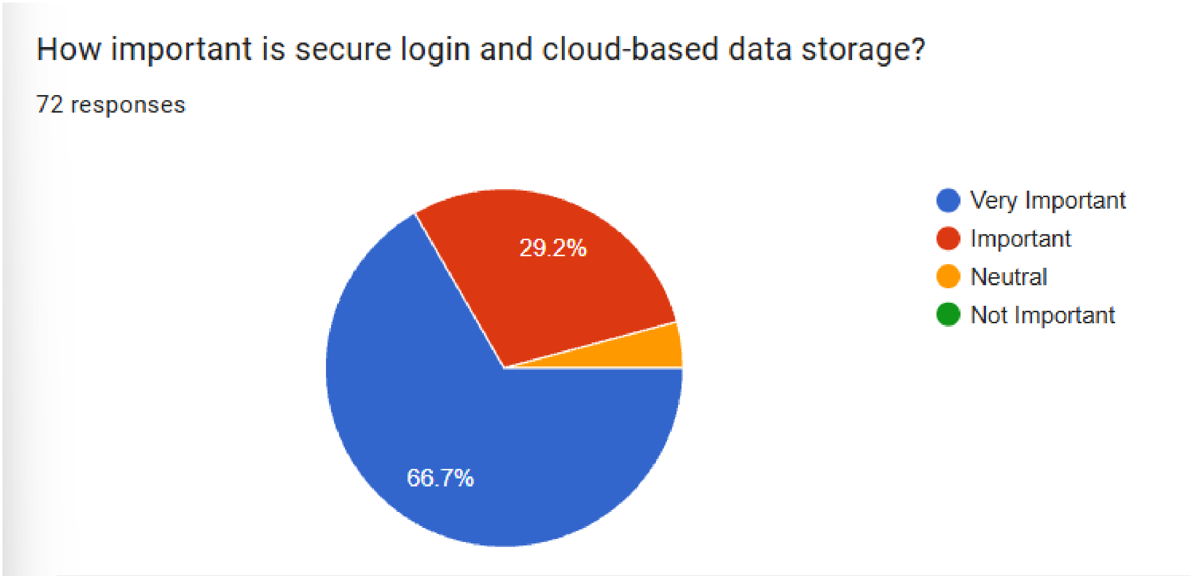


Figure 26:

How useful would AI-based event recommendations be?

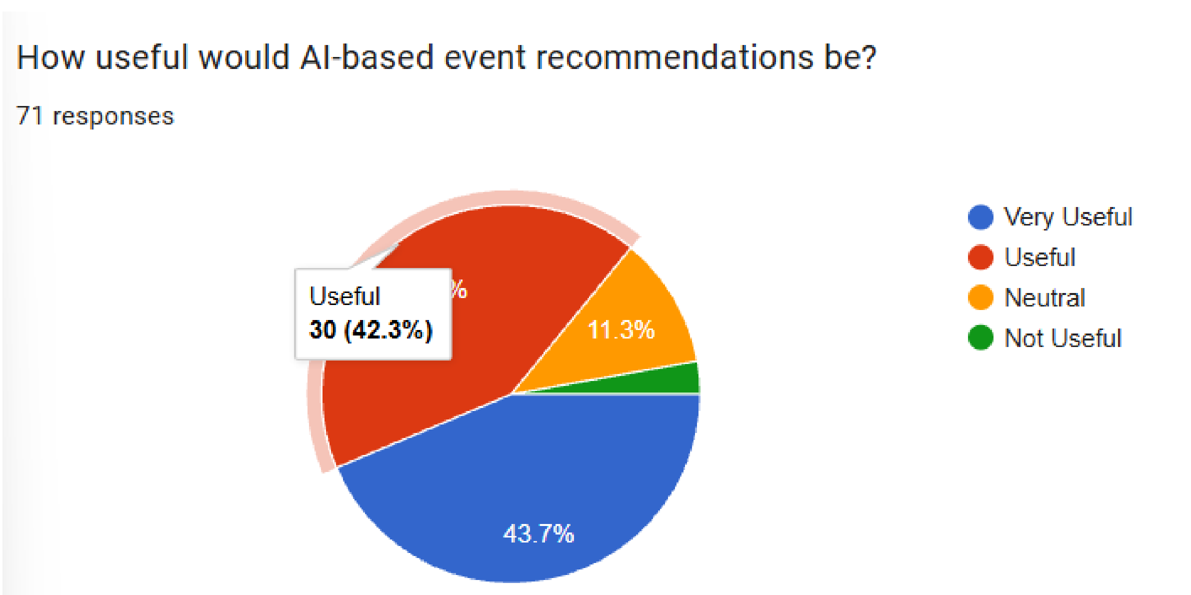


Figure 27:

How important are event reminders and notifications?

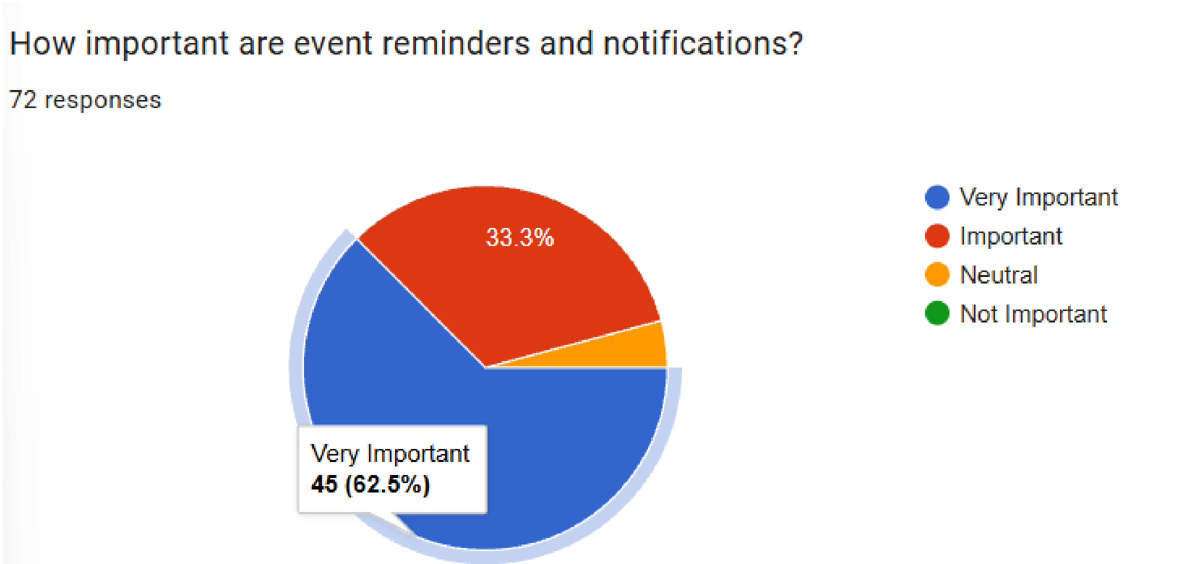


Figure 28:

How useful would QR code ticketing/check-in be?

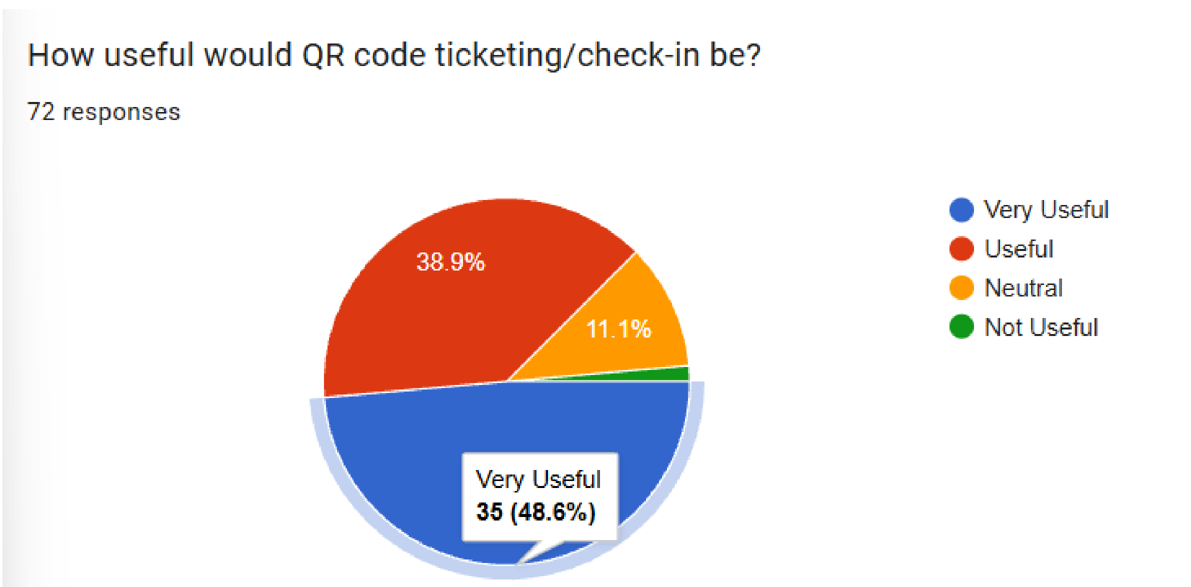


Figure 29:

How important is secure login and user access control?

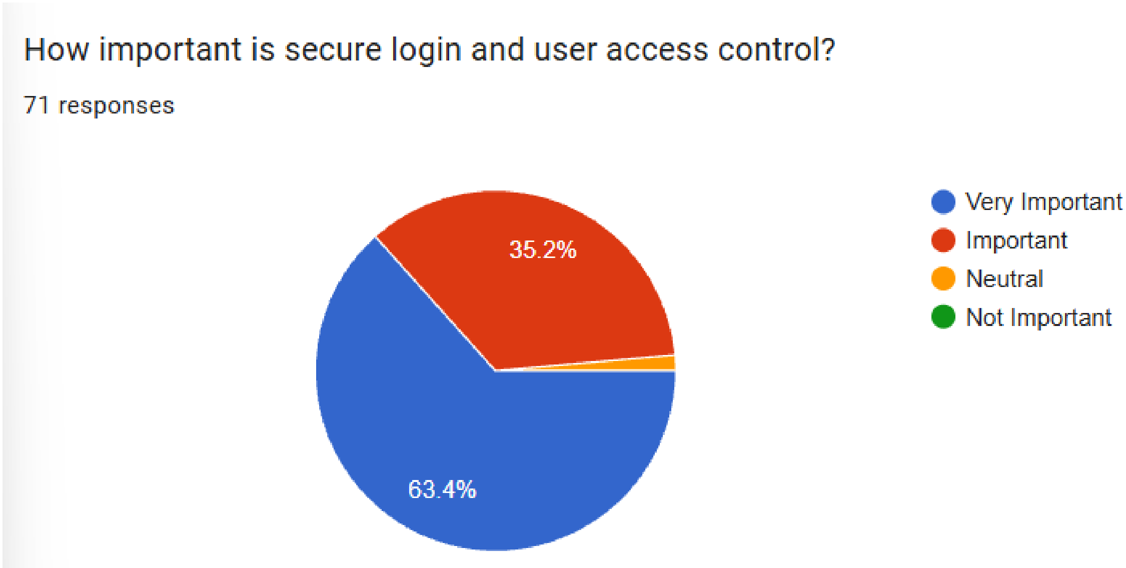


Figure 30:

How important is cloud-based data storage for event systems?

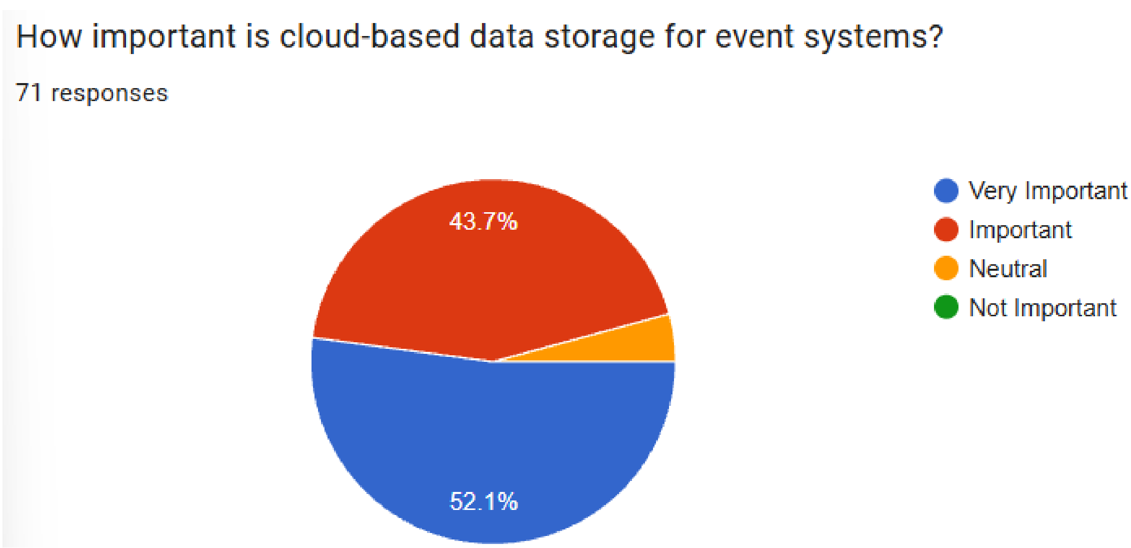


Figure 31:

How useful would an analytics dashboard be for tracking event performance?

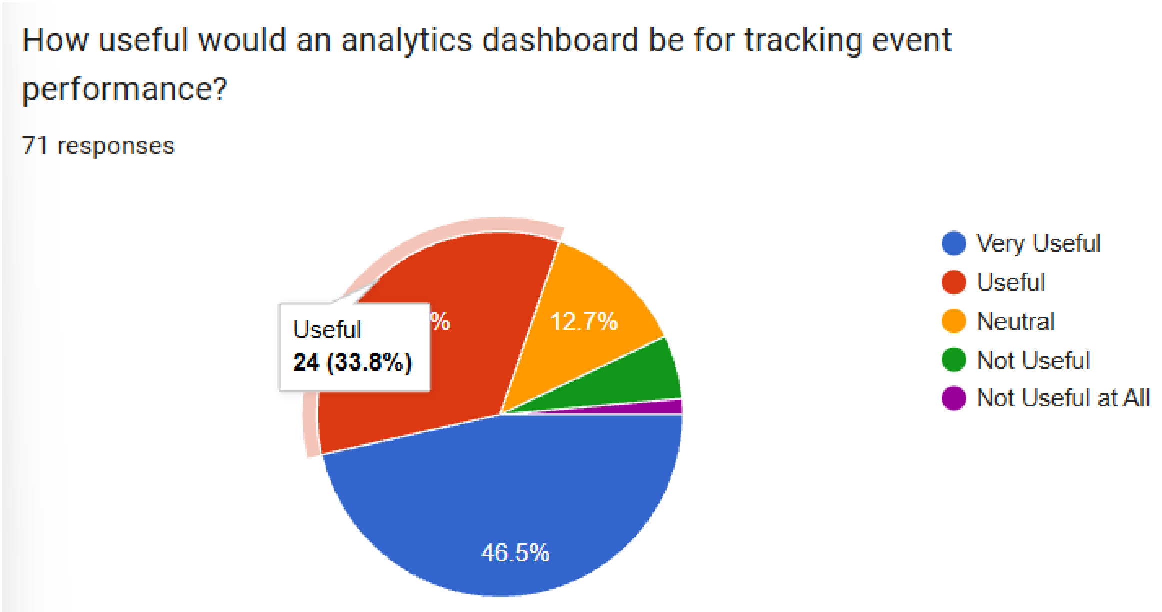


Figure 32:

Which features would you prefer in an event management system?

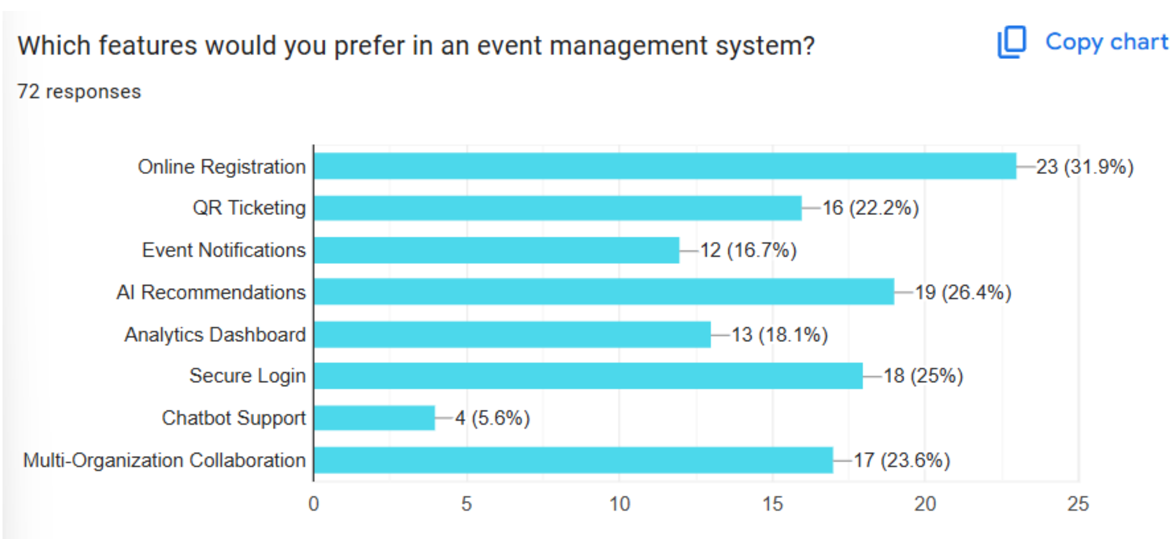


Figure 33:

Would you use an AI-powered cloud-based event management platform?